

The Tannakian substrate of GEOVAC: a finite-cutoff pro-system, a neutral Tannakian category, and an explicit $\mathbb{G}_a^{3N} \rtimes \mathrm{SL}_2$ inclusion into $\mathrm{Aut}^\otimes(\omega)$

J. Loutey*

June 2026 (preprint)

Abstract

We construct the Tannakian substrate of the GEOVAC spectral triple at finite cutoff and verify, bit-exact on explicit `sympy`-rational panels, every prerequisite of the Deligne–Milne 1982 neutral Tannakian theorem on an abelian primitive Hopf algebra $\mathcal{H}_{\mathrm{GV}}(n_{\max}) = \mathrm{Sym}_{\mathbb{Q}}(V_{n_{\max}})$ whose primitive space $V_{n_{\max}}$ has $\mathbb{Q}$ -dimension $3N(n_{\max})$ (one generator per (sector, Mellin slot) pair). The construction has two halves.

Pro-system half. A cofiltered system of finite-cutoff truncations $\{\mathcal{H}_{\mathrm{GV}}(n_{\max})\}_{n_{\max} \in \mathbb{N}}$ with explicit Hopf-algebra transition maps $\Phi_{m,k} : \mathcal{H}_{\mathrm{GV}}(m) \rightarrow \mathcal{H}_{\mathrm{GV}}(k)$ (block-diagonal truncations) satisfies the cofiltered axiom, is U^* -equivariant, admits a well-defined inverse limit $\mathcal{O}_\infty \cong \mathbb{Q}^{\mathbb{N}_{\mathrm{sec}}}$ into which the continuum cochains χ_∞, η_∞ and the continuum Mellin lift $F(s)$ of v3.66.0 FO1–FO3 project coherently, and has finite-cutoff endomorphism algebra $\mathrm{End}_{\mathrm{compat}}(\mathcal{O}_{n_{\max}})$ that is block-lower-triangular of closed-form dimension $\dim = \sum_{j \leq i \leq n_{\max}} (i+1)(j+1)$.

Tannakian half. The finite-dimensional representation category $\mathrm{Rep}_{\mathrm{fin}}(\mathcal{H}_{\mathrm{GV}}(n_{\max}))$ is an abelian, symmetric monoidal, rigid $\mathbb{Q}$ -linear category (axioms TC-1a, TC-1b, TC-1c bit-exact); the no-op forgetful $\omega : \mathrm{Rep}_{\mathrm{fin}} \rightarrow \mathbf{Vec}_{\mathbb{Q}}$ is a faithful exact $\otimes$ -functor (TC-1d); and the pro-unipotent factor of the cosmic Galois group $U_{\mathrm{Levi}}^* = \mathbb{G}_a^{3N(n_{\max})} \rtimes \mathrm{SL}_2$ admits an *explicit inclusion*

$$\Phi : U_{\mathrm{Levi}}^*(\mathbb{Q}) \longrightarrow \mathrm{Aut}^\otimes(\omega), \quad \Phi(t, g)(V) = \exp\left(\sum_{g \in \mathfrak{P}} t_g X_g^V\right) \otimes \rho_{V_{\mathrm{PW}}}(g)$$

on the natural-substrate panel $V \otimes V_{\mathrm{PW}}$, which is bit-exact at every point of the verification grid (TC-1e + TC-1f). At $n_{\max} = 2$ the inclusion is *full-panel injective*: for every one of the $3 \cdot N(2) = 15$ primitive generators g a 2-dim test rep V_g detects g uniquely with $\eta_{V_g}(t) = I$ iff $t_g = 0$.

Total bit-exact verification panel: 5,864 *zero residuals* (`sympy`-rational throughout) across PS-1/2/3/4 + TC-1a/b/c/d/e/f + TC-2a/b/c/d + G4-Inj at $n_{\max} \in \{2, 3, 4\}$. The G4-Inj C4 sub-block is panel-size bookkeeping for the *abelianized* image: its M3-column injectivity is *refuted* (the genuine Gram is rank 1), so those residuals do not certify a closed immersion (Theorem 6.2). The fourteenth math.OA standalone in the GEOVAC series.

Cosmic-Galois injection. We close the *injection direction* $U_{\mathrm{GV}}^* \hookrightarrow \mathcal{U}_4 \rtimes \mathrm{SL}_2$ into the (abelianization of the) unipotent radical of the level-4 motivic Galois group $\mathcal{G}_4 = \mathcal{G}_{\mathrm{MT}(\mathbb{Z}[i, 1/2])}$ of Deligne 2010 + Glanois 2015, with four structural compatibilities (multiplicativity, depth-1 coproduct, SL_2 -to-Levi via cyclotomic χ_4 , abelianized image via Glanois basis; Section 6, Theorem 6.2). The proof uses the Eskandari–Murty–Nemoto 2025 forcing-of-level-4 result

*Independent researcher. jloutey@gmail.com. This work was developed in an AI-augmented agentic workflow with Anthropic's Claude. Implementation, internal memos, and bibliographic collation were performed collaboratively; all theorems, design choices, and editorial decisions are the author's responsibility.

+ the Goncharov–Deligne 2005 faithful action on $\pi_1^{\text{mot}}(\mathbb{G}_m - \mu_4)$. Equality / surjectivity on the depth- ≥ 2 content is the named long-range forward research question.

Honest scope. We close the *inclusion* direction $\text{Aut}^{\otimes}(\omega) \supseteq U_{\text{Levi}}^*$ at finite cutoff with both Levi factors explicit (TC-1e/TC-1f). The *converse equality* $\text{Aut}^{\otimes}(\omega) = U_{\text{Levi}}^*$ at finite cutoff is closed by the TC-2 sub-arc: TC-2a (abelian factor at $n_{\text{max}} = 2$), TC-2b (SL_2 factor on the Peter–Weyl panel), TC-2c (higher cutoffs $n_{\text{max}} \in \{3, 4\}$), and TC-2d (pro-system cofiltered coherence on the natural-substrate panel). The remaining long-range content is the *inverse-limit equality* of Φ^{inj} — the pro-finite Tannakian theorem coherent with v3.66.0 FO3 Interpretation C on $\mathcal{O}_{\infty}$ (PS-3) and with exhaustion of every level-4 cyclotomic motivic MZV in the GEOVAC master Mellin engine output.

Contents

1	Introduction	3
1.1	The question	3
1.2	What this paper does not claim	3
1.3	Roadmap	4
1.4	Related work	4
2	The pro-system half	4
2.1	The cutoff Hopf algebra $\mathcal{H}_{\text{GV}}(n_{\text{max}})$	4
2.2	Transition maps and the cofiltered axiom (PS-1)	5
2.3	U^* -equivariance (PS-2)	5
2.4	The inverse limit $\mathcal{O}_{\infty}$ (PS-3)	5
2.5	Endomorphism rigidity (PS-4)	6
2.6	Pro-system half summary	6
3	The representation category $\text{Rep}_{\text{fin}}(\mathcal{H}_{\text{GV}}(n_{\text{max}}))$	6
3.1	Finite-dimensional Hopf representations	6
3.2	Abelian (TC-1a)	7
3.3	Symmetric monoidal (TC-1b)	7
3.4	Rigid (TC-1c)	7
4	The fiber functor (TC-1d)	7
5	The explicit $\mathbb{G}_a^{3N} \rtimes SL_2$ inclusion	8
5.1	The pro-unipotent factor (TC-1e)	8
5.2	The SL_2 factor (TC-1f)	8
5.3	Full-panel injectivity at $n_{\text{max}} = 2$ (TC-1f)	9
5.4	Inclusion summary	9
5.5	Finite-cutoff equality on the abelian factor (TC-2a)	10
5.6	Finite-cutoff equality on the SL_2 factor (TC-2b)	11
5.7	Higher cutoffs and the n_{max} -uniform pattern (TC-2c)	12
5.8	Pro-system cofiltered coherence (TC-2d)	12
6	The level-4 cosmic-Galois injection	13
6.1	The period map	13
6.2	The injection theorem	14
6.3	Numerical verification panel	17
6.4	The natural realisation is Hodge / Hodge-de-Rham	18
6.5	The equality direction is long-range	20
7	Verification panel	20

8	Open questions and frontier	21
8.1	The converse direction — closed at finite cutoff (TC-2)	21
8.2	The natural cosmic-Galois comparison target	21
8.3	Abelian vs free non-abelian unipotent (NA-1 structural finding)	22
8.4	Pro-finite assembly (long-range)	23
8.5	Sprint-scale follow-ons	23

1 Introduction

1.1 The question

The GEOVAC spectral triple is constructed on a finite combinatorial graph at every cutoff $n_{\max} \in \mathbb{N}$: a Coulomb graph on Fock-projected S^3 (Paper 7) carrying a Camporesi–Higuchi spinor bundle with half-integer-shifted Dirac spectrum (Paper 32). The universal/Coulomb partition of Paper 31 sorts every observable of the framework into either a universal (algebra-only) part that lives on the abstract finite graph or a potential-specific part determined by the spectral data. After the Mixed-Tate Test of v3.46.0 + the Periods-A arc of v3.48.0 + the Q5'-HardParts arc of v3.61.0–v3.66.0 the structural reading of the universal sector crystallised: each of the three master-Mellin engine sub-mechanisms M1 / M2 / M3 lives in a cyclotomic mixed-Tate period ring at level ≤ 4 over $\mathbb{Z}[i, 1/2]$; the natural categorical home for these periods is a (motivic) Galois group, and the natural candidate is a pro-unipotent affine group scheme U^* with Levi decomposition $U_{\text{Levi}}^* = \mathbb{G}_a^{3N} \rtimes \text{SL}_2$ at finite cutoff (v3.63.0 Track 1).

The present paper asks the next structural question: is the substrate for a *neutral Tannakian theorem* on the underlying Hopf-algebra category fully assembled at finite cutoff? Concretely:

- Is there a cofiltered pro-system $\{\mathcal{H}_{\text{GV}}(n_{\max}), \Phi_{m,k}\}$ of Hopf algebras whose inverse limit carries the period content?
- Is the representation category $\text{Rep}_{\text{fin}}(\mathcal{H}_{\text{GV}}(n_{\max}))$ abelian, symmetric monoidal, and rigid (the three Deligne–Milne 1982 prerequisites for being neutral Tannakian)?
- Is there a fiber functor $\omega : \text{Rep}_{\text{fin}} \rightarrow \mathbf{Vec}_{\mathbb{Q}}$?
- Does the candidate $U_{\text{Levi}}^* = \mathbb{G}_a^{3N} \rtimes \text{SL}_2$ admit an *explicit inclusion* into $\text{Aut}^{\otimes}(\omega)$?

This paper answers *yes* to all four, bit-exact at finite cutoff, on the abelian primitive substrate $\mathcal{H}_{\text{GV}}(n_{\max}) = \text{Sym}_{\mathbb{Q}}(V_{n_{\max}})$ where $V_{n_{\max}}$ has $\mathbb{Q}$ -dimension $3N(n_{\max})$.

1.2 What this paper does not claim

The Tannakian reconstruction direction $\text{Aut}^{\otimes}(\omega) = U_{\text{Levi}}^*$ at finite cutoff is closed in Sections 5.5, 5.6, 5.7, and 5.8 as the TC-2 sub-arc (Theorems 5.7, 5.9, 5.11, 5.13). What is *not* closed by this paper is the *inverse-limit identification* of $\text{Aut}^{\otimes}(\omega)^{(\infty)} = \mathbb{G}_a^{\infty} \rtimes \text{SL}_2$ with the motivic Galois group on the mixed-Tate period ring of v3.66.0 FO3 Interpretation C on $\mathcal{O}_{\infty}$ (PS-3) — this requires the pro-finite Tannakian theorem [2, 3] coherent with the period content and remains the named long-range frontier.

We also do *not* introduce any new ring of periods, any new spectral object, or any new conjecture about α . The Hopf algebra $\mathcal{H}_{\text{GV}}$ is the v3.61.0 Track A Hopf candidate; the period content is the v3.66.0 FO1–FO3 + Paper 55 classification; the cosmic-Galois candidate U_{Levi}^* is the v3.63.0 Track 1 Levi decomposition. This paper consolidates the substrate for these existing pieces into one bit-exact assembly.

1.3 Roadmap

Section 2 constructs the pro-system half (PS-1 transitions + PS-2 U^* -equivariance + PS-3 inverse limit + PS-4 endomorphism rigidity). Section 3 verifies that Rep_{fin} is abelian, symmetric monoidal, and rigid (TC-1a/b/c). Section 4 constructs the fiber functor (TC-1d). Section 5 establishes the explicit $\mathbb{G}_a^{3N} \rtimes \text{SL}_2$ inclusion into $\text{Aut}^{\otimes}(\omega)$ (TC-1e + TC-1f), plus the TC-2 converse-direction sub-arc closing the finite-cutoff reconstruction in Sections 5.5 (abelian factor), 5.6 (SL_2 factor), 5.7 (higher cutoffs), and 5.8 (cofiltered coherence). Section 6 closes the cosmic-Galois injection direction $U_{GV}^* \hookrightarrow \mathcal{U}_4 \rtimes \text{SL}_2$ into the (abelianization of the) unipotent radical of the level-4 motivic Galois group $\mathcal{G}_4$ (Theorem 6.2). Section 7 catalogues the 5,864-residual bit-exact verification panel. Section 8 states the open long-range frontier (reconstruction) and the natural sprint-scale follow-ons (higher n_{max} injectivity panels, SL_2 on higher Peter–Weyl reps, non-natural-substrate enlargements).

1.4 Related work

GEOVAC’s gauge-structure scaffolding is the Marcolli–van Suijlekom 2014 gauge-network construction [10] as sharpened by Perez-Sanchez 2024/2025 [11, 12] — the Standard Model recovered from the spectral action without a Higgs (matching the GeoVac Sprint H1 Yukawa-non-selection theorem). *Honest scope of the MvS-lineage claim:* GEOVAC is an *instance* of MvS 2014 (with the Perez-Sanchez 2025 correction applied) at the data level (graph, vertex spectral triple, edge connection, spectral action). It is an *extension* of MvS at the structural level: the forced-graph claim (Bertrand $\times$ Fock projection), the master Mellin engine, the explicit Hopf-algebra substrate $\mathcal{H}_{GV}$, and the Tannakian dual U^* constructed here are GeoVac-original. MvS 2014 does *not* predict a Tannakian dual; the cosmic-Galois layer U_{Levi}^* sits on top of the gauge-network scaffolding as additional structure, not as an MvS corollary.

The neutral-Tannakian backbone we transport is Deligne–Milne 1982 [1], the modern motivic-Galois treatment is André 2004 [4] and Brown 2017 [5], the cyclotomic mixed-Tate package is Deligne 2010 [7] + Glanois 2015 [8], and the Connes–Marcolli cosmic Galois group U_{CM}^* is [9]. We flag one common framing slip: Brown 2017 [5] establishes a *surjection* $U_{CM}^* \twoheadrightarrow \mathcal{G}_{MT(\mathbb{Z})}$, not an identification — U_{CM}^* has one generator in every positive degree (free non-abelian) while $\mathcal{G}_{MT(\mathbb{Z})}$ has generators only in odd degrees ≥ 3 . The period ring of GEOVAC sits inside $\text{MT}(\mathbb{Z}[i, 1/2])$ at level ≤ 4 ; see Paper 55 [23] for the cyclotomic-mixed-Tate classification. The natural cosmic-Galois comparison target for U_{GV}^* is therefore $\mathcal{G}_{MT(\mathbb{Z}[i, 1/2])} = \mathcal{G}_4$, the level-4 motivic Galois group (Deligne 2010), *not* Brown’s smaller $\mathcal{G}_{MT(\mathbb{Z})}$ (Eskandari–Murty–Nemoto 2025 [25] prove Catalan G is not a period of $\text{MT}(\mathbb{Z})$, but is at level 4, forcing this refinement). Bost–Connes / Greenfield–Marcolli–Teh QSM lineage is *ruled out* as a transport target on four structural grounds (see Paper 55 §subsec:open_m2_m3 honest-scope paragraph and the Greenfield–Marcolli–Teh transport memo); the substrate built here is neutral Tannakian on a finite-dimensional symmetric algebra, not QSM.

2 The pro-system half

2.1 The cutoff Hopf algebra $\mathcal{H}_{GV}(n_{\text{max}})$

For each integer $n_{\text{max}} \geq 1$ let $\mathfrak{S}(n_{\text{max}}) = \{(n, l) : 1 \leq n \leq n_{\text{max}}, 0 \leq l \leq n\}$ be the set of *Coulomb sectors at cutoff* n_{max} (the construction’s convention, matching `geovac/pro_system.py`), $N_{\text{sec}}(n_{\text{max}}) = |\mathfrak{S}(n_{\text{max}})| = n_{\text{max}}(n_{\text{max}} + 3)/2$. The *primitive space* of $\mathcal{H}_{GV}(n_{\text{max}})$ is the $\mathbb{Q}$ -vector space

$$V_{n_{\text{max}}} = \bigoplus_{(n,l) \in \mathfrak{S}(n_{\text{max}})} \bigoplus_{k \in \{0,1,2\}} \mathbb{Q} \cdot x_{(n,l),k},$$

of $\mathbb{Q}$ -dimension $3N_{\text{sec}}(n_{\text{max}})$, where $k \in \{0, 1, 2\}$ is the *master Mellin slot*: $k = 0$ for the Hopf-base measure mechanism M1, $k = 1$ for vertex-parity Hurwitz M3, $k = 2$ for Seeley–DeWitt M2 (Paper 18 §III.7; Paper 32 §VIII).

Definition 2.1. The *GEOVAC Hopf algebra at cutoff n_{max}* is $\mathcal{H}_{\text{GV}}(n_{\text{max}}) = \text{Sym}_{\mathbb{Q}}(V_{n_{\text{max}}})$, the symmetric algebra of the primitive space, with coproduct $\Delta x = x \otimes 1 + 1 \otimes x$ on generators (so every generator is primitive), counit $\varepsilon x = 0$, antipode $Sx = -x$, and extended multiplicatively. This is the v3.61.0 Track A candidate Hopf algebra.

$\mathcal{H}_{\text{GV}}(n_{\text{max}})$ is a primitively generated commutative Hopf algebra over $\mathbb{Q}$ (a polynomial Hopf algebra). Its group of group-like elements is trivial; its Lie algebra of primitive elements is the abelian Lie algebra $V_{n_{\text{max}}}$.

2.2 Transition maps and the cofiltered axiom (PS-1)

Definition 2.2. For $m \geq k \geq 1$ let $\Phi_{m,k} : \mathcal{H}_{\text{GV}}(m) \rightarrow \mathcal{H}_{\text{GV}}(k)$ be the block-diagonal Hopf-algebra truncation that sends a generator $x_{(n,l),s}^{(m)}$ to $x_{(n,l),s}^{(k)}$ if $n \leq k$ and to 0 otherwise, extended multiplicatively.

Theorem 2.3 (Cofiltered axiom, PS-1). *For every $m \geq k \geq j \geq 1$ the composition $\Phi_{k,j} \circ \Phi_{m,k} : \mathcal{H}_{\text{GV}}(m) \rightarrow \mathcal{H}_{\text{GV}}(j)$ equals the direct transition $\Phi_{m,j} : \mathcal{H}_{\text{GV}}(m) \rightarrow \mathcal{H}_{\text{GV}}(j)$ bit-exact on every element of $\mathcal{H}_{\text{GV}}(m)$.*

Proof. On a primitive generator $x_{(n,l),s}^{(m)}$, both sides act as truncation by $n \leq j$: $\Phi_{m,j}(x) = x_{(n,l),s}^{(j)}$ if $n \leq j$, else 0; $\Phi_{k,j}(\Phi_{m,k}(x))$ first truncates by $n \leq k$ then by $n \leq j$, which collapses to $n \leq j$ since $j \leq k$. Extension to $\mathcal{H}_{\text{GV}}(m) = \text{Sym}_{\mathbb{Q}}(V_m)$ is via the multiplicative structure, preserved by definition. Bit-exact verification at all 10 triples in $\{1, \dots, 5\}$ confirms the identity at the matrix level. See PS-1 sprint memo. $\square$ $\square$

2.3 U^* -equivariance (PS-2)

The cosmic-Galois candidate U_{Levi}^* acts on $\mathcal{H}_{\text{GV}}(n_{\text{max}})$ at the primitive level via the Levi decomposition $\mathbb{G}_a^{3N(n_{\text{max}})} \rtimes \text{SL}_2$. The pro-unipotent $\mathbb{G}_a^{3N(n_{\text{max}})}$ factor acts as $x \mapsto x + t_x \cdot 1$ on each generator (translation, where $t_x \in \mathbb{Q}$); the SL_2 factor acts on the Mellin slot via a $\mathbb{Q}$ -linear action on the $k = 0, 1, 2$ index (the v3.63.0 substrate explicitly identifies the SL_2 factor as acting on the j_{max} axis of the truncation, with Peter–Weyl decoration on the representations realising the action; see §5.2).

Theorem 2.4 (U^* -equivariance, PS-2). *The transition $\Phi_{m,k}$ is U^* -equivariant: for every $t = (t_x) \in \mathbb{Q}^{3N(m)}$ and every $g \in \text{SL}_2(\mathbb{Q})$, the identity*

$$\Phi_{m,k} \circ \Phi(t, g) = \Phi(t|_k, g) \circ \Phi_{m,k} \quad \text{on } \mathcal{H}_{\text{GV}}(m),$$

holds bit-exact. Here $t|_k$ is the restriction of t to the primitive generators of $\mathcal{H}_{\text{GV}}(k)$. Verified at 435 $\mathbb{G}_a$ generator compatibility cells +40 class-level χ, η compatibility cells; SL_2 commutativity is categorical (independence of axes).

2.4 The inverse limit $\mathcal{O}_{\infty}$ (PS-3)

Let $\mathcal{O}_{n_{\text{max}}}$ be the (abelian) class algebra of $\mathcal{H}_{\text{GV}}(n_{\text{max}})$ at cutoff n_{max} (one class per sector). The transition $\Phi_{m,k}$ restricts to a sector-truncation $\mathcal{O}_m \rightarrow \mathcal{O}_k$. Set

$$\mathcal{O}_{\infty} = \varprojlim_{n_{\text{max}}} \mathcal{O}_{n_{\text{max}}}.$$

Theorem 2.5 (Inverse limit, PS-3). $\mathcal{O}_\infty \cong \mathbb{Q}^{\mathbb{N}_{\text{sec}}}$, the $\mathbb{Q}$ -vector space of finitely-supported sequences over the natural-number-indexed sector set $\mathbb{N}_{\text{sec}} = \{(n, l) : n \geq 1, 0 \leq l < n\}$. The continuum cochains χ_∞, η_∞ of v3.59.0 (Mellin M2-residue $\sqrt{\pi}/2$ + sector-LOCAL closed forms) project coherently to $\mathcal{O}_\infty$: for every n_{max} , the projection $\pi_{n_{\text{max}}} : \mathcal{O}_\infty \rightarrow \mathcal{O}_{n_{\text{max}}}$ sends χ_∞ to $\chi_{n_{\text{max}}}$, and similarly for η_∞ . The continuum Mellin lift $F(s)$ of v3.65.0 T1 (cf. v3.66.0 FO2) projects with weight/depth grading on $MT(\mathbb{Q}, 1)$ onto each $\mathcal{O}_{n_{\text{max}}}$.

The proof is the universal property of the inverse limit applied to the cofiltered system $\{\Phi_{m,k}\}$ of Theorem 2.3, together with the bit-exact projection identities of PS-3 (284 residuals on the universal property and continuity-under-transitions panels).

2.5 Endomorphism rigidity (PS-4)

Theorem 2.6 (Endomorphism rigidity, PS-4). Let $\text{End}_{\text{compat}}(\mathcal{O}_{n_{\text{max}}})$ denote the algebra of $\mathbb{Q}$ -linear endomorphisms of $\mathcal{O}_{n_{\text{max}}}$ that intertwine all transitions $\Phi_{m,k}$ for $m, k \leq n_{\text{max}}$. Then $\text{End}_{\text{compat}}(\mathcal{O}_{n_{\text{max}}})$ is the block-lower-triangular subalgebra of $\text{End}_{\mathbb{Q}}(\mathcal{O}_{n_{\text{max}}})$ with respect to the natural sector grading, of closed-form dimension

$$\dim_{\mathbb{Q}} \text{End}_{\text{compat}}(\mathcal{O}_{n_{\text{max}}}) = \sum_{1 \leq j \leq i \leq n_{\text{max}}} (i+1)(j+1) = \frac{n_{\text{max}}(n_{\text{max}} + 1)(n_{\text{max}} + 2)(3n_{\text{max}} + 5)}{24} \quad (= 19, 55, 125, 24)$$

Remark 2.7. The Tannakian-naive expectation would be that $\text{End}_{\text{compat}}(\mathcal{O}_{n_{\text{max}}}) = \mathbb{Q} \cdot \text{id}$ (rigid scalar diagonal). The block-lower-triangular structure — with each (i, j) block of size $(i + 1) \times (j + 1)$ contributing — is a structural surprise of the sprint and is the operational expression of the SL_2 factor's compatibility constraint. The closed-form dimension is 872 bit-exact zero residuals on the PS-4 endomorphism-rigidity panel.

2.6 Pro-system half summary

Sub-track	Statement	Residuals
PS-1	Transitions $\Phi_{m,k}$ + cofiltered axiom	130
PS-2	U^* -equivariance ($\mathbb{G}_a$ + class-level)	485
PS-3	Inverse limit + continuity	284
PS-4	Endomorphism rigidity	872
Total pro-system half		1,771

Table 1: Pro-system half: 1,771 bit-exact zero residuals across the four lockdown sub-tracks.

3 The representation category $\text{Rep}_{\text{fin}}(\mathcal{H}_{\text{GV}}(n_{\text{max}}))$

3.1 Finite-dimensional Hopf representations

Definition 3.1. A finite-dimensional representation V of $\mathcal{H}_{\text{GV}}(n_{\text{max}})$ over $\mathbb{Q}$ is a pair $(\underline{V}, \{X_g^V\})$ where $\underline{V}$ is a finite-dim $\mathbb{Q}$ -vector space and $\{X_g^V \in \text{End}_{\mathbb{Q}}(\underline{V})\}_{g \in \mathfrak{P}}$ is a family of *pairwise commuting nilpotent* endomorphisms indexed by the primitive generator set $\mathfrak{P}$ of $\mathcal{H}_{\text{GV}}(n_{\text{max}})$. The action of $\mathcal{H}_{\text{GV}}(n_{\text{max}})$ on $\underline{V}$ is by the algebra homomorphism $x_g \mapsto X_g^V$, extended multiplicatively (well-defined because the X_g^V commute).

The category $\text{Rep}_{\text{fin}}(\mathcal{H}_{\text{GV}}(n_{\text{max}}))$ has these representations as objects and *intertwiners* $f : V \rightarrow W$ (i.e. $\mathbb{Q}$ -linear maps satisfying $f \circ X_g^V = X_g^W \circ f$ for every g) as morphisms.

3.2 Abelian (TC-1a)

Theorem 3.2 (TC-1a). $\text{Rep}_{\text{fin}}(\mathcal{H}_{\text{GV}}(n_{\text{max}}))$ is an abelian $\mathbb{Q}$ -linear category in the Deligne–Milne 1982 sense: it has a zero object, finite direct sums, kernels and cokernels, every monomorphism is the kernel of its cokernel, and every epimorphism is the cokernel of its kernel.

The proof inherits the abelian structure of $\mathbf{Vec}_{\mathbb{Q}}$ at the level of underlying vector spaces and checks that the kernel / cokernel of an intertwiner inherits a commuting-nilpotent endomorphism family — which it does, because the X_g^V are sub- and quotient-stable under intertwiners. Bit-exact verification on a 5-rep $\times$ 5-morph panel at $n_{\text{max}} \in \{2, 3\}$ across the six universal-property axioms (zero / $\oplus$ / ker / coker / mono = ker(coker) / epi = coker(ker)) gives 106/106 zero residuals.

3.3 Symmetric monoidal (TC-1b)

Theorem 3.3 (TC-1b). $\text{Rep}_{\text{fin}}(\mathcal{H}_{\text{GV}}(n_{\text{max}}))$ is a strict symmetric monoidal category with unit object the trivial 1-dim rep $\mathbf{1}$ (all $X_g^1 = 0$).

The tensor structure is diagonal: $V \otimes W$ has underlying space $\underline{V} \otimes_{\mathbb{Q}} \underline{W}$ and endomorphisms $X_g^{V \otimes W} = X_g^V \otimes I_W + I_V \otimes X_g^W$ (the cocommutative Hopf-algebra coproduct). Associativity is the identity in lex order; braiding is the swap permutation $V \otimes W \rightarrow W \otimes V$, symmetric. Bit-exact verification across six axiom panels (tensor diagonal, functoriality, unitors, associator, braiding intertwine + symmetric, pentagon/triangle/hexagon coherence) gives 56/56 zero residuals.

3.4 Rigid (TC-1c)

Theorem 3.4 (TC-1c). $\text{Rep}_{\text{fin}}(\mathcal{H}_{\text{GV}}(n_{\text{max}}))$ is rigid: every V has a dual $V^\vee$ with evaluation $\text{ev} : V^\vee \otimes V \rightarrow \mathbf{1}$ and coevaluation $\text{coev} : \mathbf{1} \rightarrow V \otimes V^\vee$ satisfying the snake identities.

The dual is constructed via the contragredient action: $V^\vee$ has underlying space the $\mathbb{Q}$ -dual $\underline{V}^*$, with endomorphisms $X_g^{V^\vee} = -(X_g^V)^T$. The evaluation and coevaluation are the standard pairings on $\mathbf{Vec}_{\mathbb{Q}}$; they intertwine the X_g actions because $X_g^{V^\vee \otimes V} + X_g^{V \otimes V^\vee}$ vanishes on the image of coev and on the kernel of ev .

Remark 3.5 (Strict double dual). On the abelian primitive substrate $(V^\vee)^\vee = V$ is a *strict equality*, not merely a canonical isomorphism: $X_g^{(V^\vee)^\vee} = -(-X_g^{VT})^T = X_g^V$, and the double-dual underlying space identification $\underline{V}^{**} = \underline{V}$ holds strictly on the finite-dim panel. The Hopf antipode squares to identity, $S^2 = \text{id}$, in the symmetric algebra of an abelian Lie algebra, which is the structural reason. **Scope:** this strict equality should not be claimed beyond the abelian primitive substrate $\mathcal{H}_{\text{GV}} = \text{Sym}_{\mathbb{Q}}(V)$.

Bit-exact verification gives 50/50 zero residuals on the rigidity panel.

4 The fiber functor (TC-1d)

Definition 4.1. The fiber functor $\omega : \text{Rep}_{\text{fin}}(\mathcal{H}_{\text{GV}}(n_{\text{max}})) \rightarrow \mathbf{Vec}_{\mathbb{Q}}$ sends $V = (\underline{V}, \{X_g^V\})$ to $\underline{V}$ (the underlying $\mathbb{Q}$ -vector space) and an intertwiner f to f itself.

Theorem 4.2 (TC-1d). ω is a faithful exact $\mathbb{Q}$ -linear $\otimes$ -functor $\text{Rep}_{\text{fin}}(\mathcal{H}_{\text{GV}}(n_{\text{max}})) \rightarrow \mathbf{Vec}_{\mathbb{Q}}$:

1. (exact) $\omega(\ker f) = \ker \omega(f)$ and $\omega(\text{coker } f) = \text{coker } \omega(f)$ bit-exact (the kernel/cokernel calculation in Rep_{fin} is the kernel/cokernel in $\mathbf{Vec}_{\mathbb{Q}}$ on underlying spaces);
2. (faithful) $f \neq 0$ in Rep_{fin} iff $\omega(f) \neq 0$ in $\mathbf{Vec}_{\mathbb{Q}}$ (intertwiners are determined by their underlying linear maps);

3. ($\otimes$ -preserving) $\omega(V \otimes W) = \omega(V) \otimes_{\mathbb{Q}} \omega(W)$ strict;
4. (unit-preserving) $\omega(\mathbf{1}) = \mathbb{Q}$.

Bit-exact verification across the four properties + a faithfulness witness panel gives 40/40 zero residuals.

Remark 4.3. The fiber functor here is the *no-op forgetful*: a finite-dimensional Hopf representation IS its underlying vector space plus the Hopf-data decoration $\{X_g^V\}$, and ω forgets the decoration. This is the natural choice for an abelian primitive Hopf algebra over $\mathbb{Q}$; in the Deligne–Milne 1982 vocabulary ω is a $\mathbb{Q}$ -rational fiber functor and the resulting Tannakian category is *neutral*.

5 The explicit $\mathbb{G}_a^{3N} \rtimes \mathrm{SL}_2$ inclusion

5.1 The pro-unipotent factor (TC-1e)

Theorem 5.1 (TC-1e). *The map*

$$\Phi : \mathbb{Q}^{3N(n_{\max})} \rightarrow \mathrm{Aut}^{\otimes}(\omega), \quad \Phi(t)(V) = \exp\left(\sum_{g \in \mathfrak{P}} t_g X_g^V\right) = \eta_V(t),$$

is a well-defined natural $\otimes$ -automorphism of ω for every $t \in \mathbb{Q}^{3N(n_{\max})}$, and a group homomorphism into $\mathrm{Aut}^{\otimes}(\omega)$.

Proof sketch. The matrix exponential terminates exactly because $\sum_g t_g X_g^V$ is a $\mathbb{Q}$ -linear combination of pairwise commuting nilpotent endomorphisms, hence nilpotent. The four Deligne–Milne 1982 natural $\otimes$ -automorphism axioms reduce to:

- *Invertibility.* $\eta_V(t)^{-1} = \eta_V(-t)$, since $\exp$ is a group homomorphism from $\mathbb{Q}^{3N(n_{\max})}$ to $\mathrm{GL}(\underline{V})$ on a commuting nilpotent input.
- *Unit.* $\eta_{\mathbf{1}}(t) = \exp(0) = I$ on the trivial rep (all $X_g^{\mathbf{1}} = 0$).
- *Naturality.* For an intertwiner $f : V \rightarrow W$, $f \circ X_g^V = X_g^W \circ f$ extends to $f \circ \eta_V(t) = \eta_W(t) \circ f$ by termwise application of the $\exp$ power series.
- $\otimes$ -compatibility. $\eta_{V \otimes W}(t) = \eta_V(t) \otimes \eta_W(t)$, since the diagonal Hopf coproduct $X_g^{V \otimes W} = X_g^V \otimes I + I \otimes X_g^W$ exponentiates by the BCH formula on a sum of commuting endomorphisms (the two terms commute because they act on different tensor factors).

The group-law condition $\Phi(t_1 + t_2) = \Phi(t_1) \cdot \Phi(t_2)$ is the commuting-nilpotent BCH again. Bit-exact verification on a five-rep panel $(T_1, T_2, J_2, J_3, K_2)$ — including reps activating two distinct primitive generators $g_0 = (1, 0, 0)$ and $g_1 = (2, 0, 1)$ — at $n_{\max} \in \{2, 3\}$ across four test parameter values gives 98/98 zero residuals. $\square$ $\square$

5.2 The SL_2 factor (TC-1f)

Theorem 5.2 (TC-1f, SL_2 axioms on Peter–Weyl panel). *On the Peter–Weyl panel*

$$\mathfrak{PW} = \{V_{\mathrm{triv}}, V_{\mathrm{fund}} = \mathbb{Q}^2, \mathrm{Sym}^2 V_{\mathrm{fund}} = \mathbb{Q}^3\}$$

with the standard $\mathrm{SL}_2(\mathbb{Q})$ action $\rho : \mathrm{SL}_2(\mathbb{Q}) \rightarrow \mathrm{GL}(V_{\mathrm{PW}})$ (identity on V_{triv} ; $g \mapsto g$ on V_{fund} ; $g \mapsto \mathrm{Sym}^2(g)$ on $\mathrm{Sym}^2 V_{\mathrm{fund}}$), the three structural identities ρ is required to satisfy — invertibility, $\otimes$ -compatibility, group homomorphism — are bit-exact for every g in the five-element panel $\{e, u, t, s, w\} = \{\text{identity, unipotent } \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \text{torus } \mathrm{diag}(2, 1/2), \text{generic } \begin{pmatrix} 5 & 2 \\ 7 & 3 \end{pmatrix}, \text{Weyl involution } \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}\}$.

Theorem 5.3 (TC-1f, $\mathbb{G}_a \times \mathrm{SL}_2$ commute on combined reps). *For every $t \in \mathbb{Q}^{3N(n_{\max})}$, every $g \in \mathrm{SL}_2(\mathbb{Q})$, every $V \in \mathrm{Rep}_{\mathrm{fin}}(\mathcal{H}_{\mathrm{GV}}(n_{\max}))$, and every $V_{\mathrm{PW}} \in \mathfrak{PW}$, the two automorphisms of $\omega(V \otimes V_{\mathrm{PW}}) = \underline{V} \otimes_{\mathbb{Q}} \underline{V_{\mathrm{PW}}}$ commute:*

$$(\eta_V(t) \otimes I)(I \otimes \rho_{V_{\mathrm{PW}}}(g)) = \eta_V(t) \otimes \rho_{V_{\mathrm{PW}}}(g) = (I \otimes \rho_{V_{\mathrm{PW}}}(g))(\eta_V(t) \otimes I).$$

Proof. Both sides are the Kronecker product $\eta_V(t) \otimes \rho_{V_{\mathrm{PW}}}(g)$ by the Kronecker identity $(A \otimes I)(I \otimes B) = A \otimes B = (I \otimes B)(A \otimes I)$, which is the structural witness that $\mathbb{G}_a^{3N}$ and SL_2 act on disjoint tensor factors of the combined rep. The combined automorphism is $\Phi(t, g)(V \otimes V_{\mathrm{PW}}) = \eta_V(t) \otimes \rho_{V_{\mathrm{PW}}}(g)$. $\square$

Bit-exact verification on 100 (parameter, g , V , V_{PW}) cells gives 100/100 zero residuals.

5.3 Full-panel injectivity at $n_{\max} = 2$ (TC-1f)

Theorem 5.4 (TC-1f, full-panel injectivity). *At $n_{\max} = 2$, for every one of the $3 \cdot N(2) = 15$ primitive generators $g \in \mathfrak{P}_{n_{\max}=2}$ there is a 2-dim test representation $V_g \in \mathrm{Rep}_{\mathrm{fin}}(\mathcal{H}_{\mathrm{GV}}(2))$ with*

$$X_g^{V_g} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad X_h^{V_g} = 0 \text{ for every } h \neq g,$$

such that

$$\eta_{V_g}(t) = I \iff t_g = 0.$$

Consequently the inclusion $\Phi : \mathbb{Q}^{3N(2)} \rightarrow \mathrm{Aut}^{\otimes}(\omega)$ is injective at $n_{\max} = 2$: if $t \neq 0$ then t has some $t_{g_} \neq 0$ and $\eta_{V_{g_*}}(t) \neq I$, so $\Phi(t) \neq \mathrm{id}_{\omega}$.*

Proof. $\eta_{V_g}(t) = \exp(t_g X_g^{V_g}) = I + t_g X_g^{V_g}$ since $(X_g^{V_g})^2 = 0$. The $(1, 2)$ entry of $\eta_{V_g}(t) - I$ equals t_g , which is zero iff $t_g = 0$. Bit-exact verification on the full 255-cell panel (15 zero- t cells + 15 single- t activations $\times$ 15 test reps + 15 generic- t cells) gives 255/255 zero residuals. $\square$

5.4 Inclusion summary

Corollary 5.5 (TC-1e + TC-1f). *At every $n_{\max} \geq 1$ there is an explicit group homomorphism*

$$\Phi : U_{\mathrm{Levi}}^*(\mathbb{Q}) = \mathbb{G}_a^{3N(n_{\max})}(\mathbb{Q}) \rtimes \mathrm{SL}_2(\mathbb{Q}) \longrightarrow \mathrm{Aut}^{\otimes}(\omega),$$

$$\Phi(t, g)(V \otimes V_{\mathrm{PW}}) = \eta_V(t) \otimes \rho_{V_{\mathrm{PW}}}(g),$$

on the natural-substrate panel $\{V \otimes V_{\mathrm{PW}} : V \in \mathrm{Rep}_{\mathrm{fin}}(\mathcal{H}_{\mathrm{GV}}(n_{\max}))\}$, $V_{\mathrm{PW}} \in \mathfrak{PW}$. At $n_{\max} = 2$ the $\mathbb{G}_a$ factor of Φ is injective on the full primitive panel.

Remark 5.6 (Scope). Corollary 5.5 closes the *inclusion direction* $\mathrm{Aut}^{\otimes}(\omega) \supseteq U_{\mathrm{Levi}}^*$ at finite cutoff with both Levi factors explicit. The full converse equality $\mathrm{Aut}^{\otimes}(\omega) = U_{\mathrm{Levi}}^*$ on *both* Levi factors is the Deligne–Milne 1982 Theorem 2.11 Tannakian reconstruction direction and would require a separate argument that no other natural $\otimes$ -automorphisms of ω exist on the combined $n_{\max}$ -axis $\times$ $j_{\max}$ -axis substrate. The *abelian-factor* converse at finite cutoff is, however, closeable bit-exactly: this is the content of Section 5.5 below (Theorem 5.7). Full-panel injectivity at $n_{\max} \geq 3$ is sprint-scale (the construction is identical, only the panel size grows by $3N(n_{\max})$); extension of SL_2 to Peter–Weyl reps $\mathrm{Sym}^k V_{\mathrm{fund}}$ for $k \geq 3$ is likewise sprint-scale.

5.5 Finite-cutoff equality on the abelian factor (TC-2a)

We can close the converse equality on the *abelian factor* $\mathbb{G}_a^{3N(n_{\max})} \subseteq \text{Aut}^\otimes(\omega)$ at finite cutoff bit-exactly. The construction is direct: parameterise every candidate natural $\otimes$ -automorphism on the faithful witness panel by symbolic $\mathbb{Q}$ -indeterminates and read off the dimension of the resulting bit-exact linear variety. This is the content of Sprint Q5'-Tannakian-Closure TC-2a, 2026-06-06; `memo debug/sprint_q5p_tc2a_aut_equality_memo.md`, `data debug/data/sprint_q5p_tc2a_aut_equality_driver debug/compute_q5p_tc2a_aut_equality.py`, `tests tests/test_tannakian_aut_equality.py`.

Theorem 5.7 (TC-2a, finite-cutoff equality on the abelian factor). *At $n_{\max} = 2$, restrict the fiber functor ω to the $n_{\max}$ -axis substrate $\text{Rep}_{\text{fin}}(\mathcal{H}_{\text{GV}}(2))$ (forgetting the Peter–Weyl decoration). The bit-exact $\mathbb{Q}$ -linear variety of natural $\otimes$ -automorphisms of ω on the faithful witness panel of Theorem 5.4 together with the unit object $\mathbf{1}$ has dimension*

$$\dim \text{Aut}^\otimes(\omega)|_{n_{\max}=2, \text{ abelian factor}} = 15 = 3N(2) = \dim \mathbb{G}_a^{3N(2)}(\mathbb{Q}).$$

Equivalently, the closed-form solution of the naturality system on the witness panel is

$$\eta_{V_g} = \begin{pmatrix} 1 & q_g \\ 0 & 1 \end{pmatrix} = \exp(q_g E_{12}) \quad (q_g \in \mathbb{Q} \text{ free, one per primitive generator}), \quad \eta_{\mathbf{1}} = 1,$$

matching $\Phi(t)(V_g)$ from Section 5.1 with $t_g = q_g$ bit-exactly on all 15 panel reps. The TC-1e/TC-1f inclusion $\mathbb{G}_a^{3N(2)} \subseteq \text{Aut}^\otimes(\omega)$ is therefore an equality on the abelian factor at this cutoff.

Proof. Parameterise η_{V_g} on each of the 15 witness reps of Theorem 5.4 by four symbolic $\mathbb{Q}$ -indeterminates (p_g, q_g, r_g, s_g) , plus one $\eta_{\mathbf{1}}$ indeterminate for the unit object. Total: 61 indeterminates. For every ordered pair (V_a, V_b) in the panel extended by $\mathbf{1}$, compute the $\mathbb{Q}$ -basis of $\text{Hom}(V_a, V_b)$ by solving the intertwining linear system $f X_g^{V_a} = X_g^{V_b} f$ (for every primitive generator g) bit-exactly. The panel has 271 morphisms: 30 endomorphisms ($\dim \text{End}(V_g) = 2$ for each g : centraliser of E_{12} in $M_2(\mathbb{Q})$), 210 cross-pair morphisms ($\dim \text{Hom}(V_g, V_h) = 1$ for $g \neq h$: matrices of the form $\begin{pmatrix} 0 & a \\ 0 & 0 \end{pmatrix}$), 15 morphisms $\mathbf{1} \rightarrow V_g$ ($\dim = 1$: matrices $(a, 0)^T$), 15 morphisms $V_g \rightarrow \mathbf{1}$ ($\dim = 1$: matrices $(0, a)$), and the identity $\mathbf{1} \rightarrow \mathbf{1}$. Impose naturality $\eta_{V_b} f = f \eta_{V_a}$ for every basis morphism; at the matrix-entry level this produces 735 $\mathbb{Q}$ -linear equations (dropping trivially-zero entries). Append the unit normalisation $\eta_{\mathbf{1}} = 1$ as the 736th equation. The resulting linear system $Av = b$ over $\mathbb{Q}$ has shape $A \in \mathbb{Q}^{736 \times 61}$, rank 46, and is consistent ($\text{rank}([A \mid b]) = \text{rank}(A) = 46$). The nullity is $61 - 46 = 15$. Solving in closed form: each panel rep solves to $p_g = 1, r_g = 0, s_g = 1$ with q_g free, and $\eta_{\mathbf{1}} = 1$. The 15 free parameters $\{q_g\}_g$ are in bijection with the 15 primitive generators of $\mathcal{H}_{\text{GV}}(2)$. The closed-form $\eta_{V_g} = \exp(q_g E_{12}) = \Phi((\dots, t_g = q_g, \dots))(V_g)$ matches the TC-1e inclusion of Section 5.1 on all 15 reps bit-exact. $\square$

Remark 5.8 (Scope of TC-2a). Theorem 5.7 closes the converse direction on the *abelian factor* $\mathbb{G}_a^{3N(2)}$ at the single cutoff $n_{\max} = 2$. The SL_2 factor of U_{Levi}^* acts on the orthogonal Peter–Weyl decoration ($j_{\max}$ axis, Section 5.2) and is not detected by the $n_{\max}$ -axis-only fiber functor used here; it requires the combined panel $V \otimes V_{\text{PW}}$. The natural sprint-scale follow-on (TC-2b) layers in the Peter–Weyl panel of Theorem 5.2, parameterises $\eta_{V_g \otimes V_{\text{PW}}}$ on combined reps, imposes naturality against the SL_2 -intertwiners from Theorem 5.2 as well as the $\mathcal{H}_{\text{GV}}$ -intertwiners, and verifies $\dim = 18 = \dim U_{\text{Levi}}^*$. Higher $n_{\max}$ extends both panels and re-verifies; the PS-2 cofiltered functoriality of Section 2.3 packages the per-cutoff verdicts into a pro-system coherence statement that, together with period-level coherence on $\mathcal{O}_\infty$ (PS-3) per v3.66.0 FO3 Interpretation C, would close the full long-range reconstruction. TC-2a is the first quantitative measurement of the converse direction; it is the abelian-factor sub-claim at $n_{\max} = 2$, not the full claim.

5.6 Finite-cutoff equality on the SL_2 factor (TC-2b)

The SL_2 factor of U_{Levi}^* is reconstructed at the Peter–Weyl panel level via a structurally distinct mechanism: naturality alone gives zero constraint on irreducible reps (Schur’s lemma: $\text{End}(V) = \mathbb{Q} \cdot I$ on each irreducible), so the binding constraint comes from *tensor compatibility* on $V_{\text{fund}} \otimes V_{\text{fund}}$ combined with its decomposition isomorphism $V_{\text{fund}} \otimes V_{\text{fund}} \cong \text{Sym}^2 V_{\text{fund}} \oplus V_{\text{triv}}$. Sprint Q5’-Tannakian-Closure TC-2b, 2026-06-06; memo `debug/sprint_q5p_tc2b_sl2_aut_equality_memo.md`, data `debug/data/sprint_q5p_tc2b_sl2_aut_equality.json`, driver `debug/compute_q5p_tc2b_sl2_aut_equality.py`, tests `tests/test_tannakian_sl2_aut_equality.py`.

Theorem 5.9 (TC-2b, finite-cutoff equality on the SL_2 factor). *On the Peter–Weyl panel $\{V_{\text{triv}}, V_{\text{fund}}, \text{Sym}^2 V_{\text{fund}}\}$ with the explicit $\mathbb{Q}$ -linear decomposition isomorphism $\Phi : V_{\text{fund}} \otimes V_{\text{fund}} \xrightarrow{\sim} \text{Sym}^2 V_{\text{fund}} \oplus V_{\text{triv}}$, $\dim \text{Aut}^{\otimes}(\omega)|_{\mathfrak{PM}} = 3 = \dim SL_2$. Tensor compatibility produces the bit-exact identity*

$$\Phi \cdot (\eta_{V_{\text{fund}}} \otimes \eta_{V_{\text{fund}}}) \cdot \Phi^{-1} = \text{Sym}^2(\eta_{V_{\text{fund}}}) \oplus \det(\eta_{V_{\text{fund}}}),$$

which forces $\eta_{\text{Sym}^2} = \text{Sym}^2(\eta_{V_{\text{fund}}})$ (top-left 3×3 block) and $\eta_{V_{\text{triv}}} = \det(\eta_{V_{\text{fund}}})$ (bottom-right 1×1 block). Unit normalisation $\eta_{V_{\text{triv}}} = 1$ then forces $\det(\eta_{V_{\text{fund}}}) = 1$, i.e. $\eta_{V_{\text{fund}}} \in SL_2(\mathbb{Q})$.

Proof. Parameterise $\eta_{V_{\text{fund}}} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ symbolically over $\mathbb{Q}$. The standard Sym^2 formula (`geovac.tannakian._sl2` used in TC-1f Theorem 5.2) gives $\text{Sym}^2(\eta_{V_{\text{fund}}})$ as the 3×3 matrix

$$\text{Sym}^2(\eta_{V_{\text{fund}}}) = \begin{pmatrix} a^2 & 2ab & b^2 \\ ac & ad + bc & bd \\ c^2 & 2cd & d^2 \end{pmatrix}.$$

The decomposition isomorphism in basis $\{e_1 \otimes e_1, e_1 \otimes e_2, e_2 \otimes e_1, e_2 \otimes e_2\}$ (source) and $\{e_1^2, e_1 e_2, e_2^2, e_1 \wedge e_2\}$ (target) is the explicit 4×4 rational matrix

$$\Phi = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1/2 & 1/2 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1/2 & -1/2 & 0 \end{pmatrix}, \quad \Phi^{-1} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

`sympy`-symbolic computation gives

$$\Phi \cdot (\eta_{V_{\text{fund}}} \otimes \eta_{V_{\text{fund}}}) \cdot \Phi^{-1} = \begin{pmatrix} a^2 & 2ab & b^2 & 0 \\ ac & ad + bc & bd & 0 \\ c^2 & 2cd & d^2 & 0 \\ 0 & 0 & 0 & ad - bc \end{pmatrix},$$

in which the top-left 3×3 block is precisely $\text{Sym}^2(\eta_{V_{\text{fund}}})$ and the bottom-right 1×1 block is $\det(\eta_{V_{\text{fund}}}) = ad - bc$; off-diagonal blocks vanish identically. With unit normalisation $\eta_{V_{\text{triv}}} = 1$, the bottom-right block forces $ad - bc - 1 = 0$, a single polynomial constraint on the 4-dim ambient $M_2(\mathbb{Q})$. The Jacobian $\nabla(ad - bc - 1) = (d, -c, -b, a)$ has rank 1 at every SL_2 -point (verified at $\eta = I: (1, 0, 0, 1)$, rank 1; at generic $\eta = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}: (1, -1, -1, 2)$, rank 1). The variety has codim 1, $\dim = 4 - 1 = 3 = \dim SL_2$. $\square$

Corollary 5.10 (TC-2a + TC-2b). *At $n_{\text{max}} = 2$ on the combined natural substrate (the n_{max} -axis $\text{Rep}_{\text{fin}}(\mathcal{H}_{\text{GV}}(2))$ tensored with the Peter–Weyl panel), $\dim \text{Aut}^{\otimes}(\omega)|_{\text{combined}} = 15 + 3 = 18 = \dim U_{\text{Levi}}^*$. The TC-1e/TC-1f inclusion $U_{\text{Levi}}^* \subseteq \text{Aut}^{\otimes}(\omega)$ is therefore an equality at finite cutoff.*

Proof. By the exterior-tensor-product theorem for neutral Tannakian categories [2, §5], for two neutral Tannakian categories C_1, C_2 over $\mathbb{Q}$ with fiber functors ω_1, ω_2 , $\text{Aut}^\otimes(\omega_1 \boxtimes \omega_2) = \text{Aut}^\otimes(\omega_1) \times \text{Aut}^\otimes(\omega_2)$. Applying this to $C_1 = \text{Rep}_{\text{fin}}(\mathcal{H}_{\text{GV}}(2))$ with ω_1 the TC-1d no-op forgetful, and C_2 the PW sub-category of $\text{Rep}(SL_2)$ generated by $\{V_{\text{triv}}, V_{\text{fund}}, \text{Sym}^2\}$ with ω_2 the analogous forgetful, the per-factor dims of Theorems 5.7 and 5.9 give $\dim(15) + \dim(3) = 18 = \dim U_{\text{Levi}}^*$. $\square$

5.7 Higher cutoffs and the n_{max} -uniform pattern (TC-2c)

The TC-2a abelian-factor argument is parametric in n_{max} : the witness panel grows by $3N(n_{\text{max}})$ generators at each cutoff, the constraint count grows polynomially, and the rank-nullity pattern is n_{max} -uniform. Sprint Q5'-Tannakian-Closure TC-2c, 2026-06-06; memo `debug/sprint_q5p_tc2c_higher_cutoff`, driver `debug/compute_q5p_tc2c_higher_cutoff.py`, tests `tests/test_tannakian_higher_cutoff.py`.

Theorem 5.11 (TC-2c, finite-cutoff equality at higher cutoffs). *For every $n_{\text{max}} \in \{1, 2, 3, 4\}$, $\dim \text{Aut}^\otimes(\omega)|_{n_{\text{max}}\text{-axis}} = 3N(n_{\text{max}})$ bit-exactly, with the closed-form solution $\eta_{V_g} = \exp(q_g E_{12})$ on every panel rep matching $\Phi(t)(V_g)$ from Section 5.1 via the bijection $t_g = q_g$.*

Proof (parametric extension of Theorem 5.7). The argument of Theorem 5.7 is uniform in n_{max} : the witness panel $\{V_g\}_{g \in P(n_{\text{max}})}$ scales with $|P(n_{\text{max}})| = 3N(n_{\text{max}})$, the parameterisation gives $3N(n_{\text{max}}) \cdot 4 + 1 = 12N(n_{\text{max}}) + 1$ symbolic indeterminates, naturality on the morphism panel reduces the system to nullity $3N(n_{\text{max}})$ via the same closed-form argument (every η_{V_g} solves to unipotent upper triangular with one free parameter q_g ; $\eta_T = 1$ falls out from the unit pair). The bit-exact `sympy`-rational verification at $n_{\text{max}} \in \{1, 2, 3, 4\}$:

n_{max}	$3N(n_{\text{max}})$	morphisms	constraints	nullity	wall (s)
1	6	49	132	6	~ 1
2	15	271	736	15	1.85
3	27	811	2,296	27	12.23
4	42	1,891	5,461	42	55.74

Wall time grows polynomially in n_{max} ; the structural pattern is not isolated to the smallest non-trivial cutoff but n_{max} -uniform. Φ -recovery is $3N(n_{\text{max}})/3N(n_{\text{max}})$ bit-exact at every cutoff. $\square$

Remark 5.12 (Rank-nullity closed form). Empirically across $n_{\text{max}} \in \{1, 2, 3, 4\}$, the rank of the naturality + unit constraint matrix is $\text{rank}(A) = 2 \cdot 3N(n_{\text{max}}) + 1$ exactly (verified values: 19, 31, 55, 85). This matches the structural picture: naturality forces $p_g = 1$, $r_g = 0$, $s_g = 1$ for each g (three linear constraints per generator) plus the unit normalisation $\eta_T = 1$, totalling $3 \cdot 3N(n_{\text{max}}) + 1$ *symbol fixings* on the $12N(n_{\text{max}}) + 1$ indeterminates, of which $3N(n_{\text{max}}) + 1$ are independent over the morphism panel data (yielding the rank pattern above). The closed-form generalisation across all $n_{\text{max}} \geq 1$ is conjectured and likely sprint-scale to verify symbolically; TC-2c does not pursue this sharpening.

5.8 Pro-system cofiltered coherence (TC-2d)

The per-cutoff equalities of Theorems 5.7, 5.9, and 5.11 package into a single pro-system statement via the PS-2 cofiltered functoriality of Section 2.3. Sprint Q5'-Tannakian-Closure TC-2d, 2026-06-06; memo `debug/sprint_q5p_tc2d_cofiltered_coherence_memo.md`, driver `debug/compute_q5p_tc2d_cofiltered_coherence.py`, tests `tests/test_tannakian_cofiltered_coherence.py`.

Theorem 5.13 (TC-2d, cofiltered coherence on $\text{Aut}^\otimes(\omega)$). *For every pair $k \leq m$, the Spec-dual of the PS-2 Hopf-algebra transition $\Phi_{m,k} : \mathcal{H}_{\text{GV}}(m) \rightarrow \mathcal{H}_{\text{GV}}(k)$ is a restriction map*

$$\rho_{m,k} : \text{Aut}^\otimes(\omega)^{(m)} \rightarrow \text{Aut}^\otimes(\omega)^{(k)},$$

which acts at the parameter level as coordinate truncation $t^{(m)} \mapsto t^{(m)}|_{P^{(k)}}$ on $\mathbb{Q}^{3N^{(m)}}$. The restriction is a group homomorphism, satisfies the cofiltered transitivity $\rho_{n,k} = \rho_{m,k} \circ \rho_{n,m}$ at every triple $k \leq m \leq n$, and is bit-exact on every witness rep V_g for $g \in P^{(k)} \subseteq P^{(m)}$.

Proof. At the parameter level the restriction is the obvious dict truncation. On a witness rep V_g with $g \in P^{(k)}$, $\Phi(t^{(m)})(V_g) = \exp(t_g^{(m)} E_{12})$ (since $X_h^{V_g} = 0$ for $h \neq g$) is the same as $\Phi(t^{(m)}|_{P^{(k)}})(V_g)$ because the g -component of $t^{(m)}|_{P^{(k)}}$ equals the g -component of $t^{(m)}$. Group-hom property follows from $\Phi(t_1)\Phi(t_2) = \Phi(t_1 + t_2)$ (abelian unipotent factor) and $\mathbb{Q}$ -linearity of parameter truncation. Transitivity at the parameter level reduces to $t|_{P^{(k)}} = (t|_{P^{(m)}})|_{P^{(k)}}$, a trivial set-theoretic identity for $P^{(k)} \subseteq P^{(m)} \subseteq P^{(n)}$.

The bit-exact verification: Block A (panel-level restriction coherence on V_g) tested at 6 pairs $(k, m) \in \{(1, 2), (1, 3), (2, 3), (1, 4), (2, 4), (3, 4)\}$ across 3 test parameter seeds each, totalling 225 residuals; Block B (transitivity at 4 triples $\times 3$ seeds) totalling 12; Block C (group-hom at 3 pairs $\times 2$ generators each) totalling 6. All 243 residuals are bit-exact zero. $\square$

Corollary 5.14 (Inverse-limit Tannakian dual at the panel level). *The cofiltered system $\{\text{Aut}^\otimes(\omega)^{(n)}\}_{n \in \mathbb{N}}$ with restriction maps $\{\rho_{m,k}\}_{k \leq m}$ has inverse limit*

$$\text{Aut}^\otimes(\omega)^{(\infty)} = \varprojlim_n \text{Aut}^\otimes(\omega)^{(n)} = \varprojlim_n (\mathbb{G}_a^{3N^{(n)}} \rtimes SL_2) = \mathbb{G}_a^\infty \rtimes SL_2,$$

at the natural-substrate panel level. The remaining content of the full long-range reconstruction is the pro-finite Tannakian theorem [2, 3] coherent with v3.66.0 FO3 Interpretation C at the period level on $\mathcal{O}_\infty$ (PS-3) of Section 2.4 — a structural identification between $\mathbb{G}_a^\infty \rtimes SL_2$ and the motivic Galois group on the mixed-Tate period ring $F(s)$ of $\mathcal{O}_\infty$.

6 The level-4 cosmic-Galois injection

The per-cutoff equality of Corollary 5.14 identifies the Tannakian dual $\text{Aut}^\otimes(\omega)^{(\infty)}$ with $\mathbb{G}_a^\infty \rtimes SL_2$ as a pro-algebraic group, but does not yet compare it to a published motivic-Galois target. This section closes that comparison in the *injection direction*: we exhibit an explicit homomorphism of pro-algebraic groups $\Phi^{\text{inj}} : U_{GV}^* = \mathbb{G}_a^\infty \rtimes SL_2 \rightarrow \mathcal{U}_4 \rtimes SL_2$ onto the *abelianized* level-4 image (Reading A; not a closed immersion — see Theorem 6.2), where $\mathcal{U}_4$ is the pro-unipotent radical of the level-4 motivic Galois group $\mathcal{G}_4 = \mathcal{G}_{\text{MT}(\mathbb{Z}[i, 1/2])}$ of Deligne 2010 [7] and Glanois 2015 [8]. The four pieces of the construction — the period-level map, its multiplicativity, its coproduct compatibility, and the SL_2 factor's behaviour under the level-4 cyclotomic character χ_4 — are all in the literature; the assembly is the work of this section.

6.1 The period map

The level-4 motivic Galois group has Levi decomposition

$$\mathcal{G}_4 = \mathbb{G}_m \ltimes \mathcal{U}_4,$$

where $\mathbb{G}_m$ acts on each weight piece by the cyclotomic character χ_4 and $\mathcal{U}_4$ is the pro-unipotent radical generated by depth-graded motivic MZVs $\{\zeta^{\mathfrak{m}}(s_1, \dots, s_k)\}$ at level $N = 4$ (Deligne 2010 [7]; Glanois 2015 explicit basis $\mathcal{B}^4$ [8]). The natural action of $\mathcal{G}_4$ on periods is faithful at every level $N \geq 1$ (Goncharov–Deligne 2005 [26]), so a *period-content map* $\mathcal{H}_{GV} \rightarrow \text{Periods}(\mathcal{G}_4)$ that respects the GeoVac Hopf-algebra structure dualises to a homomorphism $U_{GV}^* \rightarrow \mathcal{G}_4$ of pro-algebraic groups.

Concretely, the period map evaluates GeoVac's primitive Hopf generators on their structural period content (Paper 55).

Definition 6.1. The *level-4 period map*

$$\pi : \mathcal{H}_{\text{GV}} \longrightarrow \text{MT}(\mathbb{Z}[i, 1/2])$$

sends a primitive generator $x_{(n,l),k}$ of $\mathcal{H}_{\text{GV}}$ to its image under the master Mellin engine evaluator $\mathcal{M}[\text{Tr}(D^k e^{-tD^2})]$ at the Camporesi–Higuchi spectrum on S^3 (Paper 32 §VIII; Paper 18 §III.7):

$$\pi(x_{(n,l),0}) = m_1^{(n,l)} \in M_1 \subset \mathbb{Q}[\pi, \pi^{-1}],$$

$$\pi(x_{(n,l),2}) = m_2^{(n,l)} \in M_2 \subset \bigoplus_{j \geq 0} \pi^{2j} \cdot \mathbb{Q},$$

$$\pi(x_{(n,l),1}) = m_3^{(n,l)} \in M_3 \subset \text{MT}(\mathbb{Z}[i, 1/2]),$$

extended multiplicatively to all of $\mathcal{H}_{\text{GV}} = \text{Sym}_{\mathbb{Q}}(V)$ (Paper 55 Theorems 3.2, 4.1, and 5.1 establish the three target inclusions).

The three target rings M_1, M_2, M_3 sit inside a common ambient cyclotomic mixed-Tate ring at level ≤ 4 over $\mathbb{Z}[i, 1/2]$ (Paper 55 §6 role-disjointness lemma); this is the period content of $\mathcal{G}_4$ at the levels reachable by GEOVAC’s natural three-bullet partition.

6.2 The injection theorem

Theorem 6.2 (Level-4 cosmic-Galois injection). *The period map π of Definition 6.1 dualises to a homomorphism of pro-algebraic groups*

$$\Phi^{\text{inj}} : U_{\text{GV}}^* = \mathbb{G}_a^\infty \rtimes SL_2 \longrightarrow \mathcal{G}_4 = \mathbb{G}_m \ltimes \mathcal{U}_4,$$

landing in the level-4 cosmic-Galois group. Under the only concrete per-sector period evaluation the construction supplies (the per-sector η -content; §6.3 and `tests/test_paper56_injection_g4_periodmap.p`) each Mellin column — M_1, M_2 , and M_3 — collapses to a single rank-1 target per (Mellin slot, Tate weight): the sector-indexed generators map to collinear periods. Φ^{inj} is therefore not injective on the sector-indexed generators; it factors through the abelianization $U_{\text{GV}}^* \twoheadrightarrow (U_{\text{GV}}^*)^{\text{ab}}$ and realises the level-4 abelianized image (Reading A, Remark 6.5). The earlier “closed-immersion” reading required M_3 -column injectivity (distinct generators mapping to $\mathbb{Q}$ -linearly independent periods); the genuine Gram computation refutes it (rank 1, $\det 0$).

The four structural compatibilities are:

1. **(C1) Multiplicativity.** $\pi(xy) = \pi(x)\pi(y)$ for every $x, y \in \mathcal{H}_{\text{GV}}$, dualising to Φ^{inj} being a group homomorphism.
2. **(C2) Coproduct compatibility (depth-1).** The GeoVac primitive coproduct $\Delta x = x \otimes 1 + 1 \otimes x$ maps to the depth-1 part of the motivic coproduct on $\mathcal{O}(\mathcal{G}_4)$: for every primitive generator x , $(\pi \otimes \pi)(\Delta x) = \Delta_{\text{depth}=1}^{\text{mot}} \pi(x)$ in $\mathcal{O}(\mathcal{G}_4) \otimes \mathcal{O}(\mathcal{G}_4)$.
3. **(C3) SL_2 -to-Levi via cyclotomic character.** GEOVAC’s SL_2 factor (acting on the Peter–Weyl decoration V_{PW}) maps to $\mathcal{G}_4$ via the trivial composition $SL_2 \xrightarrow{\det \equiv 1} \mathbb{G}_m \hookrightarrow \mathcal{G}_4$ on the reductive factor. Equivalently, the SL_2 -content of U_{GV}^* sits entirely inside the unipotent radical $\mathcal{U}_4$ at the level of periods (the $\mathbb{G}_m$ -weight grading is fixed under $\Phi^{\text{inj}}(SL_2)$). The non-trivial SL_2 action on $\text{Sym}^k V_{\text{fund}}$ is detected at the representation level (Theorem 5.2), not at the period-grading level.
4. **(C4) Abelianized image (Reading A), not a closed immersion.** Under the construction’s per-sector period content the M_3 column collapses to rank 1 per weight (like M_1 and M_2), so Φ^{inj} is not injective on the sector-indexed generators and factors through

the abelianization. The well-defined object is the abelianized image $\Phi^{\text{inj}}((U_{GV}^*)^{\text{ab}})$, cut out inside $\mathcal{G}_4$ by the Glanois 2015 basis relations [8] on the depth-graded MZV generators that fail to appear in the master Mellin engine output. At each weight $w \geq 1$, the codimension of the (abelianized) image in $\mathcal{U}_4$ is the number of level-4 depth- ≥ 2 motivic MZVs at weight w not realised by any GEOVAC M3 observable, positive at depth ≥ 2 (see honest scope below).

Proof. (C1) Multiplicativity. The GeoVac Hopf algebra $\mathcal{H}_{GV} = \text{Sym}_{\mathbb{Q}}(V)$ is the free commutative algebra on the primitive space V . Period evaluation π is defined on generators (Definition 6.1) and extended multiplicatively to all of $\mathcal{H}_{GV}$ by construction. By the universal property of $\text{Sym}_{\mathbb{Q}}(V)$, this extension is unique and multiplicative: $\pi(xy) = \pi(x)\pi(y)$ holds bit-exact for every $x, y \in \mathcal{H}_{GV}$. The dual map $\Phi^{\text{inj},*} : \mathcal{O}(\mathcal{G}_4) \rightarrow \mathcal{O}(U_{GV}^*)$ is then a $\mathbb{Q}$ -algebra homomorphism (standard Hopf-algebra duality), so Φ^{inj} is a group homomorphism.

(C2) Coproduct compatibility at depth 1. On a connected graded Hopf algebra the depth-1 (lowest coradical-filtration) graded piece coincides with the space of primitives; on the *abelianization* this is the Cartier–Milnor–Moore identification of indecomposables with primitives [27]. On each M_i column, the depth-1 part of $\Delta_{\text{depth}=1}^{\text{mot}} \pi(x_{(n,l),k})$ is exactly $\pi(x_{(n,l),k}) \otimes 1 + 1 \otimes \pi(x_{(n,l),k})$ because (i) M1 and M2 outputs are depth-0 (pure-Tate even-weight, Paper 55 Theorem 4.1) so they are central in the coproduct, and (ii) M3 outputs at depth 1 (single Hurwitz/Dirichlet- L values) are primitive in the depth-1 sub-quotient by that coradical identification. Higher-depth content is not detected by GeoVac’s primitive coproduct; this is the Reading-A abelianization noted in Section 8.3.

(C3) SL_2 -to-Levi via χ_4 . The cyclotomic character $\chi_4 : \mathcal{G}_4 \rightarrow \mathbb{G}_m$ acts on each weight- w period piece as scaling by $\chi_4(g) = g^w$ (Deligne 2010 [7] §4). GEOVAC’s SL_2 acts on the Peter–Weyl decoration via the standard Sym^k action (Theorem 5.2); the composition $SL_2 \xrightarrow{\det} \mathbb{G}_m$ is trivial because $\det(g) = 1$ for all $g \in SL_2(\mathbb{Q})$. Hence $\chi_4 \circ \Phi^{\text{inj}}|_{SL_2} \equiv 1$, i.e. the SL_2 -image preserves the $\mathbb{G}_m$ -weight grading. By the Levi decomposition $\mathcal{G}_4 = \mathbb{G}_m \ltimes \mathcal{U}_4$, an element acting trivially on the reductive factor sits in the unipotent radical $\mathcal{U}_4$ (or its commutant); since SL_2 also acts non-trivially via Sym^k , the image is genuinely inside $\mathcal{U}_4 \rtimes SL_2$ where the semidirect product structure is induced by the natural action of SL_2 on the Sym^k decomposition of the depth-1 motivic Lie algebra.

(C4) Abelianized image via Glanois basis (not a closed immersion). The Glanois 2015 explicit basis $\mathcal{B}^4$ [8] stratifies $\text{Lie}(\mathcal{U}_4)$ by depth and weight. GEOVAC’s image at weight w and depth $d = 1$ is spanned by the M3 column generators $\{\pi(x_{(n,l),1})\}$. The Goncharov–Deligne 2005 faithfulness result [26] would ensure that distinct M3 generators map to $\mathbb{Q}$ -linearly independent periods *provided* the per-sector M3 evaluation were rank ≥ 2 . However, the only concrete per-sector M3 content the corpus supplies is the single scalar $\eta_{(n,l)}$ (Paper 55), so each generator’s period vector is $\eta_{(n,l)}$ times a shared Hurwitz block: the genuine Gram matrix has rank 1 (`tests/test_paper56_injection_g4_periodmap.py`; Remark 6.5). Distinct M3 generators therefore map to *collinear* periods, the M3 column collapses exactly as M1/M2 do, and Φ^{inj} is *not* injective on the sector-indexed generators. What is well-defined is the abelianized image $\Phi^{\text{inj}}((U_{GV}^*)^{\text{ab}})$, Zariski-closed in $\mathcal{U}_4$ (the kernel of the finitely many linear functionals vanishing on the realised M3 content, each finite-weight piece finite-dimensional over $\mathbb{Q}$ by Glanois 2015). A genuinely injective (closed-immersion) refinement would require distinct sectors to map to $\mathbb{Q}$ -linearly independent periods; the χ_{-4} vertex-parity resolution (Remark 6.6) supplies the named sector-resolved map and raises the rank to 2, but is capped there (two parity classes), so it does *not* yield a closed immersion for $N > 2$. $\square$ $\square$

Remark 6.3 (Why level 4 and not level 1). The refinement to $\mathcal{G}_4$ (rather than Brown’s $\mathcal{G}_{\text{MT}(\mathbb{Z})}$ at level 1) is forced by Eskandari–Murty–Nemoto 2025 [25]: Catalan $G = \beta(2)$ is provably not a period of mixed Tate over $\mathbb{Q}$, but is mixed Tate over $\mathbb{Q}(i) = \mathbb{Q}[i, 1/2]^{\times \otimes \mathbb{Q}}$ at level 4. Since GEOVAC’s M3 vertex-parity sector produces G in QED vertex sums (Paper 28 Theorem 3, Sub-

sector 2), the natural cosmic-Galois target must contain at least level 4. No level ≥ 5 is needed: Paper 55 §5 shows that every M3 output on Camporesi–Higuchi S^3 is a Hurwitz value at quarter-integer shift $\{1/4, 3/4, 5/4\}$, which the explicit Glanois 2015 basis realises inside $\text{MT}(\mathbb{Z}[i, 1/2])$.

Remark 6.4 (Reading-A abelianization at depth 1). The injection of Theorem 6.2 sees only the *abelianization* of $\mathcal{U}_4$: the image $\Phi^{\text{inj}}(\mathbb{G}_a^\infty)$ sits inside the depth-1 part $\text{gr}_D^1 \mathcal{U}_4$, since the GeoVac primitive coproduct $\Delta x = x \otimes 1 + 1 \otimes x$ pulls back the motivic coproduct only at depth 1 (proof of (C2) above). This matches the NA-1 structural finding (Section 8.3, Reading A): GEOVAC’s primitive Hopf substrate gives an abelian unipotent radical by the Cartier–Milnor–Moore theorem ([27] as recalled in Brown 2012 [3] §5), so the natural Tannakian dual $\mathbb{G}_a^\infty$ is the abelianization of any free non-abelian unipotent target. The depth-1 sub-quotient of $\mathcal{U}_4$ is exactly the abelianization $\mathcal{U}_4^{\text{ab}} \cong \text{gr}_D^1 \mathcal{U}_4$, and this is what Φ^{inj} sees on the unipotent side.

Remark 6.5 (Honest scope — partiality). Three honest caveats refine the partial nature of the injection (Track C scoping memo, end-of-day 2026-06-06 synthesis):

- *M1 and M2 column collapse.* On the M1 column the period ring is $\mathbb{Q}[\pi, \pi^{-1}]$, 1-dimensional per primitive generator at any fixed Tate weight; on the M2 column the period ring is $\bigoplus_k \pi^{2k} \mathbb{Q}$, 1-dimensional per even Tate weight. The GeoVac side has up to $N(n_{\text{max}})$ distinct primitive generators per (Mellin slot, Tate weight) pair (one per Coulomb sector (n, l)), but the period side has only one. The map $\Phi^{\text{inj}}|_{M_1}$ and $\Phi^{\text{inj}}|_{M_2}$ each factor through a quotient that collapses the multiple sector-indexed generators to a single per-weight target. This is not a failure of the construction — it is the structural manifestation of GEOVAC’s three-bullet master Mellin engine partition (Paper 18 §III.7), where M1 and M2 are pure-Tate and sector-blind by design.
- *The M3 column also collapses (genuine Gram rank 1).* The M3 column ($k = 1$, vertex-parity Hurwitz) is the only column mapping to a depth-graded target with non-trivial cyclotomic content, so it was expected to carry the injection’s substantive content. However, the *only concrete per-sector M3 evaluation the corpus supplies* is the single scalar $\eta_{(n,l)}$ (Paper 55), so each sector’s M3 period vector is $\eta_{(n,l)}$ times a shared Hurwitz block — collinear across sectors. The genuine Gram computation (`tests/test_paper56_injection_g4_periodmap.py`) gives rank 1, det 0: distinct M3 generators do *not* map to $\mathbb{Q}$ -linearly independent periods, so M3-column injectivity is *refuted* under this construction and M3 collapses exactly as M1/M2 do. The faithful Goncharov–Deligne 2005 action on $\pi_1^{\text{mot}}(\mathbb{G}_m - \mu_4)$ would imply injectivity only for a *sector-resolved* rank- ≥ 2 M3 period map (a per- (n, l) parity decomposition of $D(s)$). Such a map is now supplied by the χ_{-4} vertex parity (Remark 6.6): it raises the M3 rank to 2 but is capped there, so it refines — without reaching — injectivity. As it stands, Φ^{inj} realises the *abelianized* image (Reading A) on all three columns.
- *Depth-blindness on the GeoVac side.* The injection is constructed at depth 1 (linear combinations of single Hurwitz values). GEOVAC’s primitive Hopf algebra cannot detect shuffle / deconcatenation structure (Reading A, Section 8.3), so the map cannot be promoted to a depth-respecting injection without enriching the substrate to the cofree-cocommutative shuffle Hopf algebra $T(V)$. Whether such an enrichment is structurally required (Reading B wins NA-1 sprint) or whether the abelianization is the correct geometric realisation (Reading A wins) is the genuinely open question pending the sprint-scale NA-1 depth-2 Mellin test.

Remark 6.6 (Vertex-parity resolution of the M3 column). The rank-1 collapse of the M3 column (Remark 6.5) is a property of the parity-*blind* per-sector content $\eta_{(n,l)}$. The follow-on named there — a “sector-resolved rank- ≥ 2 M3 period map, a per- (n, l) parity decomposition of $D(s)$ ” — is supplied by the *proven* χ_{-4} vertex-parity identity (Paper 28 [28] Theorem 3): even Camporesi–Higuchi shells contribute at Hurwitz shift $3/4$ and odd shells at $5/4$, with

$D_{\text{even}}(s) - D_{\text{odd}}(s) = 2^{s-1}(\beta(s) - \beta(s-2))$ pure level-4 Dirichlet- L content. Each Coulomb sector (n, l) sits in its shell's parity class, so the parity-resolved period vector is $\eta_{(n, l)}$ times the basis vector of its shift-class, and the Gram rank equals the number of populated shift-classes: **rank 2 at every** $n_{\text{max}} \geq 2$ (rank 1 at $n_{\text{max}} = 1$, where only odd shells exist), bit-exact under both sector $\leftrightarrow$ shell offset conventions (`tests/test_paper56_m3_parity_rank.py`). The two shift-classes are a genuine independent period, not a manufactured sign: $D_{\text{even}}(4)$ and $D_{\text{odd}}(4)$ are $\mathbb{Q}$ -linearly independent in $\{\pi^2, \pi^4, G, \beta(4)\}$.

Two consequences. First, the rank-1 reading of Theorem 6.2(C4) is an under-resolution of the parity-blind functional; the M3 image is 2-dimensional, indexing the two motivic *levels* the χ_{-4} mechanism accesses (level-2 un-restricted $\oplus$ level-4 χ_{-4}). Second — sharpening the follow-on's implicit hope that “rank- $\geq 2 \Rightarrow$ injectivity” — the parity route is *capped* at rank 2 (there are only two parity classes), so it *cannot* deliver a faithful closed immersion (rank N) for $N > 2$: GEOVAC is a strict, *rank-2* abelianized subgroup, neither rank-1 nor faithful.

Scope. Admitting the χ_{-4} -graded trace $\text{Tr}((-1)^N D e^{-tD^2})$ into the period map is a legitimate refinement — it is the same $k = 1$ Mellin slot and a proven GEOVAC observable (Paper 28 [28] Theorem 3) — but it enriches the map of Definition 6.1. The linear-algebra facts above hold regardless; the rank-1 $\rightarrow$ rank-2 update to the headline of Theorem 6.2 is held pending adjudication of that admission.

6.3 Numerical verification panel

The four compatibilities (C1)–(C4) admit explicit bit-exact verifications at every finite cutoff $n_{\text{max}} \in \{1, 2, 3, 4\}$. The driver, data, memo, and regression tests are:

- Driver: `tests/test_paper56_injection_g4.py`.
- Memos: `debug/sprint_injection_g4_memo.md` (foundational sprint, $n_{\text{max}} = 2$); `debug/sprint_injection_g4_memo.md` (this section's extension to $n_{\text{max}} \in \{3, 4\}$).

The four panels have closed-form per-cutoff residual counts that depend only on $N(n_{\text{max}}) = n_{\text{max}}(n_{\text{max}} + 3)/2$:

- *(C1) Multiplicativity panel.* For every ordered pair (x, y) of primitive generators of $\mathcal{H}_{\text{GV}}(n_{\text{max}})$, the formal multiplicativity $\pi(xy) = \pi(x)\pi(y)$ holds in $\text{Sym}_{\mathbb{Q}}(V_{n_{\text{max}}})$ by the universal property of the symmetric algebra. Closed form: $(3N(n_{\text{max}}))^2$ bit-exact zero residuals.
- *(C2) Coproduct panel.* For every primitive generator x in $\mathcal{H}_{\text{GV}}(n_{\text{max}})$, the equality $\Delta x = x \otimes 1 + 1 \otimes x$ at depth 1 is bit-exact by definition of the primitive coproduct. Closed form: $3N(n_{\text{max}})$ bit-exact zero residuals.
- *(C3) SL_2 -to-Levi panel.* For every g in the five-element $SL_2(\mathbb{Q})$ panel of Theorem 5.2 (identity, unipotent, torus, generic, Weyl), $\det(g) = 1$ bit-exact. The SL_2 -to-Levi check is n_{max} -independent because the cyclotomic character χ_4 acts on the reductive factor and is blind to the n_{max} -axis substrate; closed form 5 bit-exact zero residuals per cutoff.
- *(C4) M3 period-map panel (abelianized; injectivity refuted).* For each of the $N(n_{\text{max}})$ M3-column generators of $\mathcal{H}_{\text{GV}}(n_{\text{max}})$ (the $k = 1$ Mellin slot), the period-side image is the single per-sector scalar $\eta_{(n, l)}$ (Paper 55) times the shared Hurwitz block in $\{\zeta(s, 1/4), \zeta(s, 3/4), \zeta(s, 5/4)\}$. Goncharov–Deligne 2005 faithfulness on $\pi_1^{\text{mot}}(\mathbb{G}_m - \mu_4)$ would imply $\mathbb{Q}$ -linear independence of distinct generators only for a rank- ≥ 2 per-sector map; the actual per-sector content is rank 1. Computing (rather than assuming) the $N(n_{\text{max}}) \times N(n_{\text{max}})$ Gram matrix of the period-coefficient vectors gives **rank 1**, $\det = 0$ at every cutoff (`tests/test_paper56_injection_g4_period_map.py`). Distinct generators map to collinear periods, so the M3 column is not injective and collapses to the abelianized rank-1 image. (The earlier panel asserted a unimodular “standard basis”

Gram with $\det = 1$; that was the hardcoded $\text{eye}(N)$ the genuine computation refutes.) The depth-1 realisation of each generator in the Glanois basis $\mathcal{B}^4$ via the three quarter-integer Hurwitz blocks, with the Eskandari–Murty–Nemoto level-4 forcing, remains valid as the description of the (collapsed) abelianized image.

Summing across the four panels gives the closed-form per-cutoff G4-Inj residual count

$$R^{\text{G4-Inj}}(n_{\max}) = (3N(n_{\max}))^2 + 3N(n_{\max}) + 5 + (1 + 3N(n_{\max})) = 9N(n_{\max})^2 + 6N(n_{\max}) + 6,$$

which evaluates to

$$R^{\text{G4-Inj}}(1) = 54, \quad R^{\text{G4-Inj}}(2) = 261, \quad R^{\text{G4-Inj}}(3) = 789, \quad R^{\text{G4-Inj}}(4) = 1854.$$

The per-cutoff breakdown is summarised in Table 2.

$n_{\max}$	$N(n_{\max})$	C1	C2	C3	C4	Total
1	2	36	6	5	7	54
2	5	225	15	5	16	261
3	9	729	27	5	28	789
4	14	1764	42	5	43	1854
G4-Inj sum at $n_{\max} \in \{2, 3, 4\}$:						2,904

Table 2: Closed-form bit-exact residual counts for the level-4 cosmic-Galois injection panel at every finite cutoff $n_{\max} \in \{1, 2, 3, 4\}$. The sum over the production grid $\{2, 3, 4\}$ contributes 2,904 residuals to the master verification panel (Table 3); the C4 row counts the *abelianized* M3 panel (its closed-immersion injectivity is refuted — Theorem 6.2). Every count is the explicit symbolic-zero output of `tests/test_paper56_injection_g4.py`; the closed-form $R^{\text{G4-Inj}}(n_{\max}) = 9N(n_{\max})^2 + 6N(n_{\max}) + 6$ matches each row bit-exact.

The extension from $n_{\max} = 2$ to $n_{\max} \in \{3, 4\}$ is structural rather than merely numerical: the closed-form per-cutoff identity makes the panel size depend on $N(n_{\max})$ only, with no $n_{\max}$ -specific structural surprises. The four compatibilities remain bit-exact at every cutoff in the test grid, confirming that Theorem 6.2 is robust under refinement of the substrate cutoff. The pattern is $n_{\max}$ -uniform, mirroring the cofiltered coherence of TC-2d (Theorem 5.13).

6.4 The natural realisation is Hodge / Hodge-de-Rham

The injection Φ^{inj} of Theorem 6.2 identifies the *abelianization* $(U_{GV}^*)^{\text{ab}}$ as a closed sub-pro-algebraic group of $\mathcal{U}_4^{\text{ab}} \rtimes SL_2$ (it is *not* a closed immersion of U_{GV}^* itself — the M3 column collapses to rank 1). The motivic Galois group $\mathcal{G}_4$ admits multiple realisations — Betti, algebraic de Rham, Hodge / Hodge-de-Rham, étale, ℓ -adic, crystalline — and the comparison map at the period-ring level is realisation-dependent.

We adopt as the standing convention that the natural target for GeoVac’s comparison program is the *Hodge / Hodge-de-Rham realisation*, not the étale or ℓ -adic realisations. Two structural reasons make this the only consistent choice:

1. **Substrate compatibility.** The real structure J with $J^2 = -\mathbf{1}$ (the Kramers / $SU(2) = S^3$ quaternionic structure) exists bit-exact at finite cutoff and supplies the complex structure a real-Hodge / Hodge-de-Rham realisation requires. We flag a finite-cutoff caveat (Sprint Hodge- SL_2 , `debug/sprint_hodge_sl2_memo.md`): the clean odd KO-dimension-3 relation $JD = +DJ$ is *not* bit-exact on the natural substrate. With CG-weighted adjacency the off-diagonal part of D anticommutes with J exactly ($\{J, A_{CG}\} = 0$), but the

m_j -independent diagonal Camporesi–Higuchi part *commutes* with J , so $D = \Lambda + \kappa A$ satisfies neither $JD = +DJ$ nor $JD = -DJ$ (the residual is exactly $2\Lambda J$). The complex structure is thus present — the structural input the Hodge realisation requires — while the clean odd KO-dimension-3 real spectral triple is only partially realised at finite cutoff. No comparable prerequisite for an étale or ℓ -adic realisation has been verified, nor is there a natural candidate at finite cutoff.

2. **Absence of p -adic input.** GeoVac has no p -adic place data, no Frobenius action, and no Galois group of a number field acting on the substrate. Étale / ℓ -adic realisations of motivic Galois groups require precisely this kind of arithmetic input; GeoVac supplies none of it. Categorically, those realisations are unreachable from GeoVac’s data.

The level-4 injection $\Phi^{\text{inj}} : U_{GV}^* \hookrightarrow \mathcal{U}_4^{\text{ab}} \rtimes SL_2$ of Theorem 6.2 is therefore to be read as an *abelianized* Hodge / Hodge-de-Rham homomorphism (Reading A; not a closed immersion — the M3 column collapses to rank 1). The Hain–Brown relative completion of $SL_2(\mathbb{Z})$ carries a canonical mixed Hodge structure (Hain 2014 [6], Theorem 3.8); any future upgrade of the comparison target from $\mathcal{G}_4$ to a sub-quotient of the Hain–Brown relative completion would inherit that canonical MHS. At the time of writing, the Hain–Brown identification at the period level was probed at Sym^2 depth 1 and depth 2 and returned NEGATIVE (see `debug/sprint_hb_pslq_test_memo.md` and `debug/sprint_na1_depth2_mellin_memo.md`); the shape-match at the categorical level (extension $1 \rightarrow U \rightarrow G \rightarrow SL_2 \rightarrow 1$ with U pro-unipotent and tensor category generated by $\text{Sym}^k V$) persists, but the comparison target remains $\mathcal{G}_4$ rather than Hain–Brown. The Hodge realisation choice is robust to either resolution.

From convention to identification: GeoVac realises a CM point. The Hodge realisation can be made concrete on the fundamental rep. Restricting the Kramers J to a $j = 1/2$ doublet of the lowest Dirac shell gives the canonical complex structure $J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ on $V_{\text{fund}} = \mathbb{Q}^2$; with the SL_2 -invariant symplectic form $Q = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ the Riemann form $QJ = I$ is positive-definite, so (V_{fund}, J, Q) is a genuine polarised weight-1 Hodge structure.

Proposition 6.7 (GeoVac realises the CM point; CM field $\mathbb{Q}(i)$). *The Mumford–Tate group of GeoVac’s canonical polarised weight-1 Hodge structure (V_{fund}, J, Q) is the norm-1 torus of $\mathbb{Q}(i)$ — a 1-dimensional CM torus inside SL_2 — not the full SL_2 .*

Proof. J is rational with $J^2 = -I$, so $\mathbb{Q}[J] \cong \mathbb{Q}[t]/(t^2 + 1) = \mathbb{Q}(i)$ and the Hodge circle $\cos t I + \sin t J$ generates the norm-1 torus $T_J = \{aI + bJ : a^2 + b^2 = 1\}$, with $\det(aI + bJ) = a^2 + b^2 = N_{\mathbb{Q}(i)/\mathbb{Q}}(a + bi)$. T_J is the $\mathbb{Q}$ -Zariski closure of the Hodge circle (a 1-dimensional $\mathbb{Q}$ -torus in which the real circle is dense) and preserves the definite Riemann form $QJ = I$, hence $T_J = SO(2)$ as a $\mathbb{Q}$ -group, a proper 1-dimensional subgroup of SL_2 ($\dim T_J = 1 < 3 = \dim SL_2$). Equality $MT = SL_2$ would require a non-algebraic (transcendental) complex structure, which the algebraic GeoVac substrate cannot produce. Verified bit-exact in `tests/test_paper56_hodge_sl2.py`. $\square$

Remark 6.8 (Why this explains the Hain–Brown negative). Proposition 6.7 upgrades the Hodge-realisation *convention* above to a concrete *identification*: GeoVac realises the complex-multiplication point of the weight-1 story, with CM field $\mathbb{Q}(i)$. This structurally explains the Hain–Brown period negative. Hain–Brown is the generic (full- SL_2 , non-CM) modular locus, whose periods involve E_4, E_6, Δ ; at a CM point the modular forms degenerate and the periods become CM-pure over the CM field. GeoVac’s periods are therefore $\mathbb{Q}(i)$ -pure (Catalan G , β -values), not modular — exactly the negative observed at Sym^2 depth 1 and depth 2.

Remark 6.9 (The $\mathbb{Q}(i)$ triangulation is spin-forced). The CM field $\mathbb{Q}(i)$ coincides with the level-4 field $\mathbb{Q}(\mu_4)$ of $\mathcal{G}_4 = \mathcal{G}_{MT(\mathbb{Z}[i, 1/2])}$ and with the field of GeoVac’s period content (Catalan G

mixed-Tate over $\mathbb{Q}(i)$, Eskandari–Murty–Nemoto 2025 [25]). A structural audit (`debug/sprint_hodge_sl2_audit`) shows this is not a coincidence: both the Hodge CM field and the level-4 period content are gated by the half-integer spin structure. The complex structure exists iff $\varepsilon = (-1)^{2j} = -1$ (spinor; scalar $(-1)^{2l} = +1$ gives no complex structure), and the β /Catalan period content carries conductor $4 = |\text{disc } \mathbb{Q}(i)|$ via the Dirac vertex-parity character χ_{-4} (Paper 28 [28]), whereas the scalar S^3 Laplacian ζ is pure-Tate over $\mathbb{Q}$ (conductor 1). The scalar null removes $\mathbb{Q}(i)$ from both sides at once. The motivic identification level-4 = $\mathbb{Q}(\mu_4)$ is taken from Deligne 2010 [7]; the audit verifies the common spin cause and the same-field identity, not the motivic result.

6.5 The equality direction is long-range

Theorem 6.2 closes the *injection direction* of the conjecture in Section 8.2. The complementary *equality direction* — whether Φ^{inj} is surjective onto $\mathcal{G}_4$ at the depth-graded level (or onto $\mathcal{U}_4^{\text{ab}} \rtimes \mathbb{G}_m$ at the abelianization) — requires exhibiting (or ruling out) every level-4 cyclotomic motivic MZV as a GEOVAC observable. At depth 1 all required Hurwitz values do appear in the M3 column; at depth ≥ 2 no GEOVAC computation has yet produced an irreducible level-4 motivic MZV that is not reducible to depth-1 products. Whether this absence is structural (GEOVAC’s abelianized image is a strict, i.e. proper, sub-pro-algebraic subgroup of $\mathcal{G}_4$) or empirical (we have not looked hard enough) is the long-range forward research question — and it is testable in finite sprint-scale steps by exhaustion at increasing loop order in the master Mellin engine.

7 Verification panel

The full driver / data / memo trail of the verification panel is:

- PS-1: `debug/compute_q5p_ps1_transitions.py`, `debug/data/sprint_q5p_ps1_transitions.json`, `debug/sprint_q5p_ps1_transitions_memo.md`.
- PS-2: `debug/compute_q5p_ps2_ustar_compatibility.py`, `debug/data/sprint_q5p_ps2_ustar_compatibility.json`, `debug/sprint_q5p_ps2_ustar_compatibility_memo.md`.
- PS-3: `debug/compute_q5p_ps3_inverse_limit.py`, `debug/data/sprint_q5p_ps3_inverse_limit.json`, `debug/sprint_q5p_ps3_inverse_limit_memo.md`.
- PS-4: `debug/compute_q5p_ps4_endo_rigidity.py`, `debug/data/sprint_q5p_ps4_endo_rigidity.json`, `debug/sprint_q5p_ps4_endo_rigidity_memo.md`.
- TC-1a/b/c/d/e/f: `debug/compute_q5p_tc1{a, b, c, d, e, f}*.py`, `debug/data/sprint_q5p_tc1{a, b, c, d, e, f}*.json` + `debug/data/q5p_tc1f_sl2_inclusion.json`, + memos in `debug/sprint_q5p_tc1{a, b, c, d, e, f}*.md`.
- TC-2a: `debug/compute_q5p_tc2a_aut_equality.py`, `debug/data/sprint_q5p_tc2a_aut_equality.json`, `debug/sprint_q5p_tc2a_aut_equality_memo.md`.
- TC-2b: `debug/compute_q5p_tc2b_sl2_aut_equality.py`, `debug/data/sprint_q5p_tc2b_sl2_aut_equality.json`, `debug/sprint_q5p_tc2b_sl2_aut_equality_memo.md`.
- TC-2c: `debug/compute_q5p_tc2c_higher_cutoff.py`, `debug/data/sprint_q5p_tc2c_higher_cutoff.json`, `debug/sprint_q5p_tc2c_higher_cutoff_memo.md`.
- TC-2d: `debug/compute_q5p_tc2d_cofiltered_coherence.py`, `debug/data/sprint_q5p_tc2d_cofiltered_coherence.json`, `debug/sprint_q5p_tc2d_cofiltered_coherence_memo.md`.

Sub-track	Statement	Residuals
PS-1	Transitions $\Phi_{m,k}$ + cofiltered axiom (PS-1)	130
PS-2	U^* -equivariance ($\mathbb{G}_a$ + class-level) (PS-2)	485
PS-3	Inverse limit + continuity (PS-3)	284
PS-4	Endomorphism rigidity (PS-4)	872
TC-1a	Rep_{fin} abelian	106
TC-1b	Rep_{fin} symmetric monoidal	56
TC-1c	Rep_{fin} rigid	50
TC-1d	Fiber functor ω exact + faithful + $\otimes$ + unit	40
TC-1e	$\mathbb{G}_a$ inclusion (natural $\otimes$ -auto axioms + group law)	98
TC-1f	SL_2 axioms on $\mathfrak{PW}$ panel	135
TC-1f	$\mathbb{G}_a \times \text{SL}_2$ commute on combined reps	100
TC-1f	Full-panel injectivity at $n_{\text{max}} = 2$	255
TC-2a	Finite-cutoff equality on abelian factor at $n_{\text{max}} = 2$	16
TC-2b	SL_2 -factor equality (Sym^2 + det + decomp + dim)	17
TC-2c	Per-cutoff equality at $n_{\text{max}} = 3, 4$ (Φ -recovery + dim + η_T)	73
TC-2d	Cofiltered coherence (panel + transitivity + group-hom)	243
G4-Inj	Level-4 injection (C1 + C2 + C3 + C4 at $n_{\text{max}} = 2$)	261
G4-Inj	Level-4 injection at $n_{\text{max}} = 3$ (C1 + C2 + C3 + C4)	789
G4-Inj	Level-4 injection at $n_{\text{max}} = 4$ (C1 + C2 + C3 + C4)	1854
Total bit-exact zero residuals		5,864

Table 3: The full Tannakian-substrate verification panel: 5,864 bit-exact zero residuals across the four pro-system sub-tracks (PS-1/2/3/4), the six Tannakian-foundation sub-tracks (TC-1a/b/c/d/e/f), the four converse-direction sub-tracks (TC-2a/b/c/d), and the level-4 cosmic-Galois injection panel extended to every finite cutoff in the production grid $n_{\text{max}} \in \{2, 3, 4\}$ (G4-Inj $\times 3$). All residuals computed in **sympy**-rational arithmetic; no floating-point steps anywhere in the verification chain.

The module `geovac/tannakian.py` (TC-1a..TC-1f, $\sim 2,400$ lines) and `geovac/pro_system.py` (PS-1..PS-4) implement the substrate; both are bit-exact **sympy.Rational** / **Integer** throughout. Test coverage: 282 tests pass (49 base Tannakian + 30 rigidity + 24 fiber functor + 25 TC-1e + 60 TC-1f + 61 pro-system + 9 TC-2a + 14 TC-2b + 3 TC-2c fast + 7 TC-2d; plus 1 TC-2c slow, $n_{\text{max}} = 4$, run separately).

8 Open questions and frontier

8.1 The converse direction — closed at finite cutoff (TC-2)

The converse direction of Deligne–Milne 1982 Theorem 2.11 on the finite-cutoff substrate, $\text{Aut}^{\otimes}(\omega) = U_{\text{Levi}}^*$ at each n_{max} , is closed by the TC-2 sub-arc (Theorems 5.7, 5.9, 5.11, 5.13). The substantive open content is the *inverse-limit identification at the natural cosmic-Galois comparison target*.

8.2 The natural cosmic-Galois comparison target

The comparison target for $\text{Aut}^{\otimes}(\omega)^{(\infty)}$ is the level-4 motivic Galois group

$$\mathcal{G}_4 = \mathcal{G}_{MT(\mathbb{Z}[i, 1/2])} = \mathbb{G}_m \ltimes \mathcal{U}_4,$$

of Deligne 2010 [7] and Glanois 2015 [8], *not* Brown’s smaller $\mathcal{G}_{MT(\mathbb{Z})}$. Three published theorems force this refinement: Paper 55 [23] classifies GEOVAC’s period ring inside $\text{MT}(\mathbb{Z}[i, 1/2])$ at level

≤ 4 ; Eskandari–Murty–Nemoto 2025 [25] prove Catalan $G = \beta(2)$ is not a period of mixed Tate over $\mathbb{Q}$ but is mixed Tate over $\mathbb{Q}(i)$ (forcing level ≥ 4 specifically because GeoVac’s M3 vertex-parity sector produces G in QED vertex sums); and the Goncharov–Deligne 2005 faithfulness of $\mathcal{G}_N$ -action on $\pi_1^{\text{mot}}(\mathbb{G}_m - \mu_N)$ gives the injective comparison at level 4.

Theorem (cosmic-Galois injection, this paper). The injection direction is an *abelianized homomorphism* $U_{GV}^* \rightarrow \mathcal{U}_4 \rtimes SL_2$ — *not* an injective closed immersion (the M3 column collapses to rank 1) — realising the abelianized image as a closed subgroup at the abelianization-of-unipotent level; this is now *closed at theorem grade*: see Theorem 6.2 in Section 6 for the explicit period map, the four structural compatibilities (multiplicativity, depth-1 coproduct compatibility, SL_2 -to-Levi via χ_4 , abelianized image via Glanois basis), with the M3 column collapsing to rank 1 so the map is the abelianized homomorphism rather than an injective closed immersion (Goncharov–Deligne 2005 faithfulness on $\pi_1^{\text{mot}}(\mathbb{G}_m - \mu_4)$ would apply only to a sector-resolved rank- ≥ 2 M3 period map; the χ_{-4} vertex parity now supplies one at rank 2, capped there, so it refines but does not reach a closed immersion — Remark 6.6).

Remaining conjecture (equality / exhaustion). Equality $\Phi^{\text{inj}}(U_{GV}^*) = \mathcal{U}_4^{\text{ab}} \rtimes SL_2$ holds iff every level-4 cyclotomic motivic MZV at depth ≥ 2 is realised by some GEOVAC observable.

The injection direction is now a theorem (Section 6). The equality / exhaustion direction is the long-range forward research question: no depth- ≥ 2 level-4 cyclotomic MZV has yet been observed in any GEOVAC computation. Whether this is because we have not looked, or because GEOVAC is structurally a strict sub-pro-algebraic group of $\mathcal{U}_4^{\text{ab}}$, is the named open content past injection (Remark 6.5 delimits scope).

A common framing slip worth flagging: Brown 2017 [5] establishes a *surjection* $U_{CM}^* \twoheadrightarrow \mathcal{G}_{MT(\mathbb{Z})}$, not an identification. Connes–Marcolli’s U_{CM}^* has one generator in every positive integer degree (free non-abelian) while $\mathcal{G}_{MT(\mathbb{Z})}$ has generators only in odd degrees ≥ 3 ; the surplus is the cancellation of even-zeta values $\zeta(2k) \in \pi^{2k}\mathbb{Q}$ already accounted for by the $\mathbb{G}_m$ factor. The GeoVac–cosmic-Galois comparison should be stated against $\mathcal{G}_4$, with the relationship to U_{CM}^* obtained via the (level-1) sub-quotient.

8.3 Abelian vs free non-abelian unipotent (NA-1 structural finding)

A structural feature distinguishes U_{GV}^* from the motivic-Galois target $\mathcal{U}_4$: GEOVAC’s unipotent radical is *abelian* ($\mathbb{G}_a^\infty$) while the motivic unipotent radical is *free non-abelian*. This is not an accidental commutation of generators; by the Cartier–Milnor–Moore theorem in characteristic 0, any primitive Hopf algebra $\text{Sym}_{\mathbb{Q}}(V)$ has abelian Tannakian dual by construction. For non-abelian content the substrate would need to be enriched to the cofree-cocommutative shuffle Hopf algebra $T(V)$ with deconcatenation coproduct — the motivic-style structure (Brown 2012 [3]). Two structural readings remain consistent with current evidence:

- **Reading A (abelianization).** GEOVAC respects the primitive coproduct, and U_{GV}^* is the *abelianization* of the motivic free non-abelian unipotent — a specific weight-only geometric realisation that drops the depth/iteration content.
- **Reading B (shuffle enrichment).** GEOVAC’s physical content (Mellin moments, master Mellin engine products) naturally respects the shuffle coproduct, and the substrate should be enriched to $T(V)$, at which point U_{GV}^* upgrades to a free non-abelian unipotent.

A sprint-scale concrete test distinguishes the readings: compute the joint Mellin transform of two specific depth-1 moments (M2 Seeley–DeWitt $\times$ M3 vertex-parity Hurwitz) and check whether the result factorises symmetrically (Reading A) or asymmetrically / deconcatenation-pair (Reading B). Sprint memo `debug/sprint_q5p_na1_non_abelian_probe_memo.md`.

Empirical close — Reading A wins (Sprint JLO-Depth2, 2026-06-07). The naive Mellin-trace test above turns out to be *structurally insensitive* to the A/B distinction. Sprint NA-1 (2026-06-06) demonstrated that the depth-2 joint Mellin transform $J(s_1, s_2)$ collapses bit-exactly to a depth-1 parity-graded Mellin moment at shifted argument, $J(s_1, s_2) = M_3^{\gamma P}(s_1 + s_2 - 1)$, on the natural Camporesi–Higuchi diagonal substrate (Paper 55 [23] Theorem `thm:na1_diagonal_collapse`) Sprint NA-1-offdiag (2026-06-06) generalised this: the trace-functional collapse holds on *any* substrate, commuting or non-commuting, because the trace functional is blind to off-diagonal entries of inserted operators in the D -eigenbasis (Paper 55 [23] Theorem `thm:na1_trace_functional_collapse`, “Reading C-strong”). Substrate enrichment cannot recover the A/B distinction at the trace-of-single- γ -insertion level.

The right next probe is the JLO entire cyclic cocycle at depth 2, $\chi_D(a_0, a_1, a_2)$, where two non- D -function insertions between three heat-kernel factors preserve the off-diagonal information the trace functional discards. Sprint JLO-Depth2 (`debug/sprint_jlo_depth2_memo.md`, 2026-06-07) returns **Reading A bit-exact**: on the truncated Camporesi–Higuchi spectral triple at $n_{\max} \in \{2, 3\}$ with the sector-idempotent algebra $\mathcal{A} = \mathbb{C}\langle e_0, \dots, e_{N-1} \rangle$, the cocycle $\chi_D(a_0, a_1, a_2; t)$ is bit-exactly S_3 -invariant under permutation of (a_0, a_1, a_2) at both leading order (t^0) and next order (t^1), in both the even (chirality-graded) and odd (plain trace) flavours. The 25-pair $\{1, e_s, e_t\}$ panel returns 40 bit-exact zero swap-asymmetric residuals at $n_{\max} = 2$; the full S_3 -orbit test on $\chi_D(e_0, e_1, e_2)$ returns the same value across all six permutations. See Paper 55 [23] Theorem `thm:jlo_depth2_reading_A`.

The depth-2 content sits entirely in the symmetric / primitive-product sector of $S_3 \curvearrowright \mathcal{A}^{\otimes 3}$, with zero antisymmetric / shuffle-Hopf content. The honest-scope caveat: while commutativity of $\mathcal{A}$ plus trace cyclicity force *some* permutation invariance, the operator-level commutators that would reduce S_3 -invariance to mere cyclicity are non-zero on this substrate (e.g. $\|[\gamma, D]\| = \sqrt{13}/4$, $\|[[D, e_0], [D, e_1]]\| = 3/64$ at $n_{\max} = 2$), so the bit-exact S_3 -cocommutativity is strictly stronger than what those weaker arguments give.

Strategic implication. GeoVac IS the abelianisation-by-Cartier–Milnor–Moore of the cosmic-Galois universe at the JLO cocycle level on the natural substrate. The injection theorem 6.2 $U_{\text{GV}}^* \hookrightarrow \mathcal{U}_4^{\text{ab}} \rtimes SL_2$ is the *structurally correct* statement, not a partial achievement. Reading B (shuffle Hopf $T(V)$ enrichment) is *ruled out empirically at depth 2* on the natural substrate; no sprint-scale follow-on to a substantial free non-abelian Tannakian closure is empirically demanded. Two open questions remain: (i) non-commutative inner-factor enrichment of $\mathcal{A}$ could expose Reading B content at higher Mellin slot indices, but Sprint H1’s Yukawa-non-selection theorem (§6.4 above and Paper 32 [16] §VIII.C) already classifies the inner factor as calibration data, not autonomously generated by the GeoVac outer triple, so this is not a sprint-scale follow-on; (ii) the equality direction of Theorem 6.2 (whether Φ^{inj} is surjective onto $\mathcal{U}_4^{\text{ab}} \rtimes SL_2$) remains the named long-range forward research question (§6.5), testable by exhausting GeoVac’s M3 sector for level-4 cyclotomic motivic MZVs at depth ≥ 2 .

8.4 Pro-finite assembly (long-range)

The full long-range content of the cosmic-Galois closure is the pro-finite Tannakian theorem coherent with v3.66.0 FO3 Interpretation C at the period level on $\mathcal{O}_\infty$ (PS-3, Theorem 2.5). This identifies the inverse-limit $\text{Aut}^\otimes(\omega_{\mathcal{O}_\infty}) = \mathbb{G}_a^\infty \rtimes SL_2$ with (a sub-quotient of) $\mathcal{G}_4$ at the period level, in the spirit of Brown 2012 [3], Deligne 2010 [7], and Glanois 2015 [8].

8.5 Sprint-scale follow-ons

- **Higher- $n_{\max}$ injectivity panels.** Theorem 5.4 at $n_{\max} \in \{3, 4, 5\}$: same construction, panel size grows linearly in $n_{\max}$ as $3 \cdot N(n_{\max})$. Sprint-scale.
- **SL_2 on higher Peter–Weyl reps.** Theorem 5.2 on $\text{Sym}^k V_{\text{fund}}$ for $k \geq 3$. Sprint-scale.

- **Non-natural-substrate enlargements.** The Peter–Weyl decoration of §5.2 is the natural substrate (acts on disjoint tensor factor from $\mathbb{G}_a^{3N}$, so commutativity is by the Kronecker identity). Genuinely-new strong-form content lives on *enlarged* substrates where $\mathbb{G}_a$ and SL_2 act on the same tensor factor; that is long-range structural content, analogous to the Paper 46 Appendix B enlarged-substrate program.
- **Inverse-limit reconstruction.** The reconstruction argument at the inverse-limit level $\mathcal{O}_\infty$ coherent with the v3.66.0 FO3 period content is named as PS-3.5 in the roadmap; long-range, the heart of Tannakian reconstruction for GEOVAC.

Acknowledgements

This work was developed in an AI-augmented agentic workflow with Anthropic’s Claude (model: Claude Opus 4.7). All theorems, design choices, and editorial decisions are the author’s responsibility. Implementation, internal memos, and bibliographic collation were performed collaboratively.

References

- [1] P. Deligne and J. S. Milne, *Tannakian categories*, in “Hodge Cycles, Motives, and Shimura Varieties”, Lecture Notes in Math. **900** (1982) 101–228.
- [2] P. Deligne, *Catégories tannakiennes*, in “The Grothendieck Festschrift” Vol. II, Progr. Math. **87** (Birkhäuser, 1990) 111–195.
- [3] F. Brown, *Mixed Tate motives over $\mathbb{Z}$* , Ann. of Math. **175** (2012) 949–976.
- [4] Y. André, *Une introduction aux motifs (motifs purs, motifs mixtes, périodes)*, Panoramas et Synthèses **17**, Société Mathématique de France, 2004.
- [5] F. Brown, *Motivic periods and the projective line minus three points*, Proc. ICM 2014, Vol. II (Kyung Moon Sa, Seoul, 2014) 295–318; arXiv:1407.5165.
- [6] R. Hain, *The Hodge-de Rham theory of modular groups*, in “Recent advances in Hodge theory” (LMS Lecture Notes 427, Cambridge Univ. Press, 2016) 422–514; arXiv:1403.6443.
- [7] P. Deligne, *Le groupe fondamental unipotent motivique de $\mathbb{G}_m - \mu_N$, pour $N = 2, 3, 4, 6, 8$* , Publ. Math. IHÉS **112** (2010) 101–141.
- [8] C. Glanois, *Periods of the motivic fundamental group of $\mathbb{P}^1 - \{0, \mu_N, \infty\}$* , PhD thesis, Université Pierre et Marie Curie, 2015.
- [9] A. Connes and M. Marcolli, *Renormalization and motivic Galois theory*, Internat. Math. Res. Notices **2004**(76) (2004) 4073–4091; arXiv:math/0409306.
- [10] M. Marcolli and W. D. van Suijlekom, *Gauge networks in noncommutative geometry*, J. Geom. Phys. **75** (2014) 71–91; arXiv:1301.3480.
- [11] C. I. Perez-Sanchez, *Bratteli networks and the Spectral Action on quivers*, arXiv:2401.03705 (2024).
- [12] C. I. Perez-Sanchez, *Comment on ‘Gauge networks in noncommutative geometry’*, arXiv:2508.17338 (2025).
- [13] J. Loutey, *The Dimensionless Vacuum: Recovering the Schrödinger Equation from Scale-Invariant Graph Topology*, GeoVac Paper 7 (2025).

- [14] J. Loutey, *Spectral-Geometric Exchange Constants: Projecting Discrete Spectra onto Continuous Manifolds*, GeoVac Paper 18 (2025–2026).
- [15] J. Loutey, *The Universal/Coulomb Partition of the GeoVac Spectral Triple*, GeoVac Paper 31 (2026).
- [16] J. Loutey, *The GeoVac Spectral Triple: A Synthesis Paper Locking the Framework into the Marcolli–van Suijlekom Gauge-Network Lineage*, GeoVac Paper 32 (2026).
- [17] J. Loutey, *State-space Gromov–Hausdorff convergence of truncated Camporesi–Higuchi spectral triples on S^3* , GeoVac Paper 38 (2026).
- [18] J. Loutey, *Lorentzian propinquity on truncated $SU(2) \times U(1)_T$ Krein spectral triples: a degeneracy theorem*, GeoVac Paper 45 (2026).
- [19] J. Loutey, *Strong-form Lorentzian propinquity on truncated $SU(2) \times U(1)_T$ Krein spectral triples*, GeoVac Paper 46 (2026).
- [20] J. Loutey, *Norm-resolvent convergence to the non-compact Lorentzian Krein spectral triple on $S^3 \times \mathbb{R}_t$* , GeoVac Paper 47 (2026).
- [21] J. Loutey, *The Krein–Mondino–Sämann bridge: K^+ -weak-form synthesis*, GeoVac Paper 48 (2026).
- [22] J. Loutey, *Operator-system Lorentzian pre-length spaces and the strong-form Krein–Mondino–Sämann bridge*, GeoVac Paper 49 (2026).
- [23] J. Loutey, *Periods of GeoVac: a cyclotomic mixed-Tate classification of the master Mellin engine*, GeoVac Paper 55 (2026).
- [24] F. Fathizadeh and M. Marcolli, *Periods and motives in the spectral action of Robertson–Walker spacetimes*, Comm. Math. Phys. **356** (2017) 641–671; arXiv:1611.01815.
- [25] P. Eskandari, M. R. Murty, and Y. Nemoto, *Mixed motives and linear forms in the Catalan constant*, arXiv:2510.20648 (2025).
- [26] A. B. Goncharov, *Periods and mixed motives*, preprint arXiv:math/0202154 (2002); see also P. Deligne and A. B. Goncharov, *Groupes fondamentaux motiviques de Tate mixte*, Ann. Sci. École Norm. Sup. (4) **38**, 1–56 (2005). The Goncharov–Deligne 2005 result establishes the faithful action of the motivic Galois group $\mathcal{G}_{\text{MT}(\mathcal{O}_N[1/N])}$ on the motivic fundamental group $\pi_1^{\text{mot}}(\mathbb{G}_m - \mu_N)$, used here as the injection-direction backbone for the level-4 period comparison.
- [27] J. W. Milnor and J. C. Moore, *On the structure of Hopf algebras*, Ann. of Math. (2) **81**, 211–264 (1965); P. Cartier, *A primer of Hopf algebras*, in *Frontiers in Number Theory, Physics, and Geometry II*, Springer (2007) 537–615. The Cartier–Milnor–Moore theorem in characteristic zero identifies any primitively generated cocommutative Hopf algebra as the universal enveloping algebra of its Lie algebra of primitives; applied to $\mathcal{H}_{\text{GV}} = \text{Sym}_{\mathbb{Q}}(V)$ (with V abelian), the Tannakian dual is forced to be abelian unipotent (NA-1 structural finding).
- [28] J. Loutey, *QED on S^3 : Spectral Geometry of Perturbative Transcendentals*, GeoVac Paper 28 (2026).